



Assessing the Relative Quality of Rock Bolt Grouting in Weak Rock Mass Based on In Situ and Numerical Pull-Out Tests

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Abstract

Grouted rock bolts anchor various in-tunnel fixtures, such as jet fans, TBM launch frames, heavy-lifting pullies or chain blocks, and abutment and corbel supports. In such applications, rock bolts are mainly subject to direct axial tensile loads rather than combined axial shear as and when working as reinforcement elements. The axial tensile load transfer between bolt and rock depends on the strength and stiffness of the grout–rock or grout–bolt interface parameters, which strongly depend on grouting quality along the bonded length. Assessing grouting quality in situ conditions is challenging, and a limited number of studies have focused on this. This paper proposes a potential method to evaluate the relative quality of grouting. The proposal is based on the limited in situ pull-out tests on short and long-grouted rock bolts compared with three-dimensional numerical simulations using the Itasca 3DEC program. Therefore, further investigation and studies are required before it is taken into practical application. However, the calibrated rock–bolt interface shear stiffness correlated well with grouting quality in this instance. It was, therefore, used as an input to assess the relative quality of grouting. A threshold value of the relative quality of grouting may be proposed as an acceptance criterion for in-situ pull-out tests. Finally, the present study investigated the sensitivity of the rock–bolt interface shear stiffness and bond strength to the load transfer between the bolt and the rock. This investigation was done through an actual application where grouted rock bolts were used to anchor lifting points for a large-scale energy project.

Highlights

- Detailed review of the load transfer mechanism for rock bolts in situ pull-out conditions.
- Development of an approach to assess the grouting quality for rock bolt in situ pull-out tests.
- Proposal of additional acceptance criteria for analysing rock bolt in situ pull-out test results.
- Sample sensitivity analysis of the grouted rock–bolt interface properties for an actual application.

Keywords Rock bolt · Grouting · In situ pull-out test · Numerical modelling · In-tunnel anchorage

1 Introduction

Among available rock bolt options, fully grouted rock bolts are widely used in civil tunnel projects due to their simplicity and fast installation, enhanced durability when double corrosion protected, ease of post-grouting when combined with mechanical anchor and high tensile capacity (Martín et al. 2013; Jin-feng and Peng-hao 2019; Li and Høien 2023). They are used as rock reinforcements for rock and or as anchorage elements (Kaiser et al. 1996; Hutchinson and Diederichs 1996; Windsor 1997; Li 2017). For example, horizontally bedded rock can be reinforced with grouted

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rock bolts to reduce bedding slippage, creating a thick roof-reinforced beam that minimises roof deflection (e.g., Pells 2002; Oliveira and Diederichs 2017; Thirukumaran and Oliveira 2023). Grouted rock bolts can anchor locally unstable roof wedges/blocks (Hutchinson and Diederichs 1996; Windsor 1997; Li 2017). In addition, they can anchor internal fixtures in tunnels or underground structures such as Jet fans, TBM launch frames, heavy-lifting pullies or chain blocks, abutments and corbel supports. It should be noted that while rock bolts providing reinforcement are subject to both tensile and shear loads, they mainly experience tensile loads when working as anchorage elements. This paper will only discuss cases where the grouted rock bolt is subject primarily to axial tensile loads.

Numerous laboratory pull-out studies have been carried out to investigate the axial load transfer mechanism between rock bolts and grout (Farmer 1975; Benmokrane et al. 1995; Kılıç et al. 2003; Martín et al. 2013; Ma et al. 2013; Thenevin et al. 2017; Teymen and Kılıç 2018; Jinfeng and Peng-hao 2019; Li and Høien 2023; Nourizadeh et al. 2025). In many of these studies, the failure mechanism of the grouted rock bolt was forced to either shear failure between the interface of grout and rock bolt or the structural tensile failure of the rock bolt itself (e.g., Benmokrane et al. 1995; Kılıç et al. 2003; Martín et al. 2013; Ma et al. 2013; Thenevin et al. 2017). These failure mechanisms simulated in laboratory conditions may be more suitable for installing grouted rock bolts in a high strength and competent rock mass. In these cases, the shear strength of the interface between grout and rock is higher than the shear strength of the interface between the rock bolt and the grout due to a higher compressive strength of the rock than the grout material. However, in practice, the more predominant failure mechanism of grouted rock bolts in weak rock masses subjected to axial tensile load is the debonding between the interface of the grout and the rock (Farmer 1975; Cai et al. 2004). Limited attention has been given to this failure mechanism in laboratory conditions.

Although laboratory studies have advanced our understanding of the load transfer between the grout and the rock bolt, these may not accurately replicate the load transfer between grout and rock in *in situ* conditions. This limitation may be associated with the fact that the bond between the grout and the rock may be affected by discrete geological features, such as bedding or joint planes (with or without soft infill material) and weak or sheared zones, and, consequently, alter the load transfer mechanisms (Li and Stillborg 1999; Zou 2004; Fu et al. 2024). It is important to note that the quality of grouting along the drill hole can have a significant impact on the strength and stiffness of the bond between the rock and grout or rock bolt and grout, as well as the load transfer mechanism (Zou 2004; Villaescusa et al. 2008; Ma et al. 2017; Wu et al. 2019; Bačić et al. 2020; Paredes et al.

2024; Fu et al. 2024). Therefore, most tunnelling projects specify, pre-production and production tests, which are conducted on the grouted rock bolts to assess the expected bond strength and rock bolt failure mechanism for each anticipated ground condition before and during the installation of the rock bolt support. In some countries, the pre and production tests are called investigation/suitability and acceptance tests, respectively (BS EN and ISO 22477–5 2018; ASTM 2013; Lardner et al. 1985). The investigation test refers to an *in situ* pull-out test that measures the ultimate bond strength between grout and rock or grout and bolt, whichever fails first. Therefore, suitable free (i.e., unbonded) and fixed (i.e., bonded) bolt lengths and test loads should be pre-calculated for each anticipated ground condition in order to guarantee actual testing of the bond. Typically, shorter bonded/fixed lengths (between 0.5 m and 1 m) and a test load of 80% of the tensile strength of the steel rebar are used in the investigation tests. In some instances, the investigation tests also use a longer bonded/fixed length with a higher test load to assess the quality of installation and grouting. Acceptance tests use rock bolts with typical design fixed (i.e. bonded) and free lengths (i.e. unbonded) through *in situ* pull-out tests, which measure rock bolt performance against the design load, which often is less than the yield tensile capacity of the steel rebar. When a rock bolt fails during an acceptance test, a replacement is installed and testing is carried out on three neighbouring rock bolts.

Assessing the quality of the *in situ* pull-out test results in major tunnelling projects within a limited time frame is challenging due to program and production constraints. The quality of these results may be affected by poor grouting, poor procedures and testing (Villaescusa et al. 2008; Ma et al. 2017; Salcher and Bertuzzi 2018). A limited number of existing studies have been focused on assessing grouting quality in long bolts, say greater than 3 m, and its impact on the strength and stiffness of grout–rock and bolt–grout interfaces and load transfer between bolt and grout or rock (Villaescusa et al. 2008; Ma et al. 2017; Salcher and Bertuzzi 2018; Wu et al. 2019; Bačić et al. 2020; Lee and Yu 2024; Sun et al. 2024). These studies were either destructive tests (e.g. the pull-out or push tests on the overcoring *in situ* rock bolt samples) (Villaescusa et al. 2008; Ma et al. 2017; Salcher and Bertuzzi 2018) and nondestructive tests (e.g., wave reflections) (Beard et al. 2003; Hartman et al. 2010; Song et al. 2017; Zou et al. 2010; Cui and Zou 2012; Wu et al. 2019; Bačić et al. 2020; Lee and Yu 2024; Sun et al. 2024). In other studies with destructive tests, rock bolts instrumented with strain gauges (e.g. Farmer 1975) or optical fibre sensors (e.g. Forbes et al. 2020) were used in *in situ* pull-out tests to study the stress and strain distribution along the rock bolts with various bond lengths. Although these tests could have provided valuable insights, their primary focus was not to investigate the grouting quality along bolt length.

Nondestructive tests require considerable time, competent personnel, continuous calibration, and verification and have limited applicability in in situ conditions. As such, they may not be the best option for program-driven large-scale tunnelling projects. Therefore, this paper proposes an alternative method to assess grouting quality. The approach was investigated using a numerical back analysis of in situ pull-out test results for grouted CT rock bolts based on the actual data from a testing campaign. The 3DEC numerical code was used to calibrate these in situ pull tests. The relative grouting quality was assessed based on the back-calculated grout shear stiffness for shorter and longer bonded lengths. Finally, the sensitivity of the rock–bolt interface parameters was demonstrated through an application of this data to the design of anchoring the heavy-loaded lifting points from rock bolts at an intersection of a mined and a TBM outfall tunnel for an energy project.

2 Behaviour of Grouted Rock Bolt Under in Situ Pull-Out Conditions

In most cases, rock bolts are installed during excavation as temporary or permanent means of support. From a wide variety of products, grouted bolts are often used in systematic support systems. Moreover, in some instances, rock bolts are partially grouted (i.e., fixed or bonded length) with a defined free (unbonded) length, depending on the application (Fig. 1a). For example, rock bolts may be as primary means of anchorage, such as for jet fan support, lifting point anchors, partial restraint for other structures and underground foundations or abutments anchorage. High axial tensile forces may be anticipated on the rock bolts in these applications. If these high bolt forces mobilise too close to the tunnel opening, they may overload the

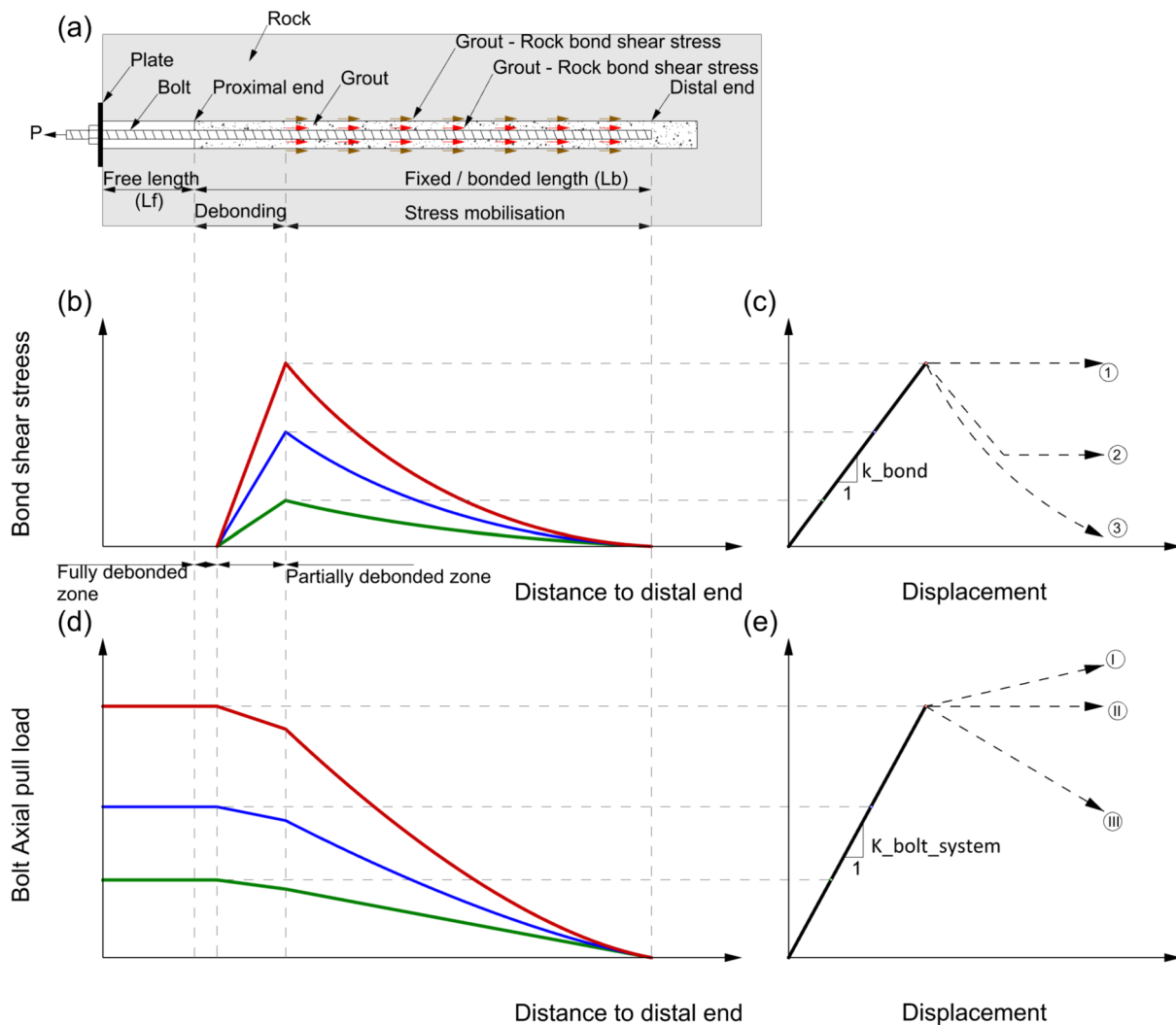


Fig. 1 Schematic diagram of stress/load transformation mechanism of grouted rock bolt subject to axial tensile load (i.e. pull out load (P)). k_{bond} is the interface shear stiffness and K_{bolt_system} is the axial stiffness of the grouted rock bolt system

existing tunnel lining and/or rock wedges and cause rock cone pull-out failure and/or grout debonding. Therefore, allowing a free/unbonded length near the rock bolt head may be beneficial as it minimises the force transferred to the rock within this length.

2.1 Force Mobilisation at the Grouted Rock–Bolt Interface

The stress/force mobilisation along the grouted rock bolts under in situ conditions is complex compared to laboratory conditions (Li and Høyen 2023) and limited data is available (Liu et al. 2017; Salcher and Bertuzzi 2018; Hadjigeorgiou and Tomasone 2018; Forbes et al. 2020). Among these, the studies from Forbes et al. (2020) and Liu et al. (2017) reported detailed stress mobilisation along bolts. A few other studies (e.g. Sun 1983 and Cai et al. 2004) focused on the grouted rock bolts installed as tunnel support, which may experience both shear and axial forces during tunnel relaxation. Most recently, Li and Høyen (2023) conducted numerous pull-out tests on various bond lengths relative to the critical bond length at a laboratory scale to simulate the anticipated behaviour of grouted bolts under in situ pull-out conditions. Li and Høyen (2023) define the critical bond as a minimum bonded length required to prevent bolt slip. Based on these existing studies, it can be confirmed that the distribution of shear stress between the interface bolt and grout or the grout and rock (i.e., bond shear stress) along the bond length is not uniform (Fig. 1b). As shown in Fig. 1b, a partial or complete debonding occurs at the start of the bond length (i.e. the proximal end), and the bond shear stress mobilises along the rest of the bond length. For each applied bolt axial force, the bond shear stress linearly increases from a minimum or residual value to a peak at the end of the partially debonded zone. Then, it exponentially decreases toward the end of the bond length (i.e., the bolt distal end; see Fig. 1a). The mobilisation of bond shear stress increases with the bolt axial force until the bond shear stress reaches its maximum value. This distribution of bond shear stress is more applicable where the bond length is equal to or more than the critical bond length, and it is more likely to take place for in situ pull-out as the lengths of grouted rock bolts installed in tunnel support are larger than the critical bond length (Li and Høyen 2023).

2.2 Shear Behaviour of the Grouted Rock–Bolt Interface

As shown in Fig. 1c, the bond shear stress linearly mobilises with the bolt axial displacement until it reaches a maximum or peak bond shear stress near the end of the debonding zone. During this pre-peak stage, the

mobilisation of the bond shear stress upon displacements is governed by the interface shear stiffness (k_{bond}) between bolt and grout or grout and rock. The existing studies show that three groups can characterise post-peak bond shear stress–displacement behaviour: They are:

- Type 1—Fully plastic (i.e. bi-linear model) which may be applicable for longer bond lengths in a weak rock mass or a competent or high-strength rock mass with a lower grout–bolt bond (Cai 2004; Liu et al. 2017; Li 2024);
- Type 2—linear strain softening with a constant residual strength (tri-linear model) which may be applicable for longer or shorter bond length in a competent or high-strength rock mass and grout (Benmokrane et al. 1995; Ren et al. 2010; Martín et al. 2013; Ma et al. 2016; Chen and Li 2022);
- Type 3—nonlinear strain softening with a very low or zero residual strength (nonlinear model), respectively and these post-peak behaviours may apply to shorter bond lengths with high-strength rock mass/host material and grout, typically at laboratory conditions (Ma et al. 2013; Jin-feng and Peng-hao 2019).

Similar to rock joint post-peak shear behaviour at different scales (Bandis et al. 1981), the post-peak bond shear stress–displacement response between grout–rock or grout–bolt interface may change from brittle (tri and non-linear models) to plastic (bi-linear model) with increasing bond length. The post-peak shear behaviour is influenced by factors such as the strength and roughness of the interface between the grout and bolt or grout and rock, as well as confining stress (i.e. the stress normal to the interface). The interfaces with more roughness and strength exhibit brittle post-peak behaviour compared to those with less roughness and strength, which show more plastic post-peak behaviour (Bandis et al. 1981). Except perhaps in highly fractured rocks with controlled borehole drilling, the surface roughness of drill boreholes for rock bolt installation can be considered to be between planar and slightly rough (i.e. Joint Roughness Coefficient (JRC) < 4), indicating that if slippage occurs between the interface of rock and grout, the shear behaviour of that interface is more likely to approach Type 1 (bi-linear) under in situ conditions. It should be noted that most in situ pull tests have long bond lengths, and their test loads are within the pre-peak stage of bond shear stress mobilisation unless the test fails with poor bond quality. On the other hand, if the shear failure occurs at the interface of the grout and the bolt, a reduction of peak strength may occur during the post-peak stage, depending on the roughness of the bolt circumference. Post-peak bond shear stress displacement is likely closer to Type 2 in this case.

2.3 Axial Behaviour of Rock Bolt

As shown in Fig. 1d, the bolt axial pull load distribution along the bond length integrates bond shear stress to bond length. Therefore, it is a nonlinear distribution. The bolt axial pull load is maximum near the proximal end and decreases to zero at the distal end. The bolt axial pull load is constant within its free length. Similar to bond shear stress–displacement behaviour, bolt axial pull load linearly increases with displacement until it reaches the yield tensile capacity of the steel rebar. The rate of increase in axial pull load with displacement is governed by an equivalent axial stiffness of the full system ($K_{\text{bolt_system}}$), including the bolt elongation and the interface shear stiffness between bolt and grout or grout and rock (k_{bond}) within the pre-yield stage. The post-yield or peak behaviour depends on the mobilisation of bond shear stress and ductility of the bolt (Fig. 1e) (Li and Høien 2023). For example, if the bond length is much greater than the critical bond length, post-yield behaviour is expected to be slightly strain-hardening until the bolt reaches its break load (Type I in Fig. 1e). If the bond length is equivalent to the critical bond, the post-yield behaviour is expected to be Type II. In contrast, if the bond length is less than the critical bond length, the pull load may not reach the bolt's yield load, and the post-peak behaviour is likely to be type III or II.

3 In Situ Pull-Out Test

For the testing discussed in this paper, in situ pull-out tests were carried out as a part of design requirements for in-tunnel lifting point anchors at an intersection of a mined outfall and TBM tunnels. Each lifting point was designed to be supported by two closed-spaced grouted CT rock bolts and a lifting plate. These lifting points were used to lift 12 tons of weight of steel beams. The CT rock bolts were installed at the side walls and shoulders of an outfall tunnel excavated

in a predominantly mudstone (85%) interbedded with limestone (15%) rock mass formation (UCS $\approx$ 12 MPa and GSI $\approx$ 55). Pull-out tests were used to assess the ultimate bond capacity and quality of grouting and bolt installation.

3.1 Bolt and Grout Properties

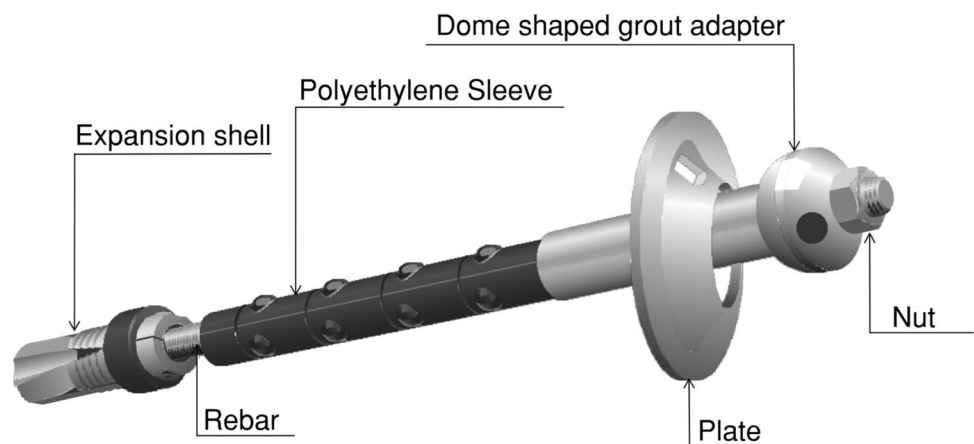
The rock bolts used in the in situ pull-out tests were CT rock bolts, which are typically installed for both temporary and permanent anchorage and reinforcement support with an optional mechanical expansion shell for pre-tension/active support. The main components of the CT rock bolt are shown in Fig. 2. The grouting adapter provides an option for post-grouting. The polyethylene sheet protects against corrosion along the bolt length. The CT rock bolt can be automatically or manually installed. The minimum yield and ultimate tensile strength of the rebar of the CT bolt are 600 MPa and 730 MPa, respectively. The diameter of the rebar of the CT bolt is around 29.73 mm whereas the length of the CT bolt varies from 3.4 m to 4.5 m. The diameter of the borehole is 68 mm. The general parameters of the CT rock bolt are given in Table 1.

The grouting material used in the testing was the chemical Mineral Bolt ULTRA SLOW from the DSI underground. This is a two-component (A and B) high-strength silicate resin that can be injected through the grouting adapter in the same manner as cementitious grout. The components A and B can be pumped using a two-component pump, as

Table 1 CT rock bolt size and strength parameters

Parameters	Value
Rebar core diameter (mm)	29.73
Elastic modulus (GPa)	200
Minimum yield strength (MPa)	600
Minimum ultimate strength (MPa)	730
Rupture strain (%)	7.5

Fig. 2 A typical CT rock bolt components



commonly used in the mining and tunnelling industry, with a volumetric ratio of 1:1. This grout material is most suitable for weak and heavily fractured rock masses where the possibility of water ingress and gas leakage is anticipated. This grout material cures rapidly to reach a compressive strength of over 50 MPa within one day. The compressive test of this resin grout material shows that the average uniaxial compressive strength and modulus of elasticity at 24 h are 52.7 MPa and 0.409 GPa, respectively.

3.2 Test Equipment

The in situ pull-out test arrangement includes a hydraulic cylinder connected to a pump and pressure gauge, a digital displacement gauge and a loading frame (Fig. 3). The extension threaded bar was fastened to the bolt head via a coupler. The hydraulic cylinder and loading frame were slid onto the extension threaded bar. The head of the hydraulic cylinder was tightened up with the extension threaded bar. The digital displacement gauge was attached between the hydraulic cylinder and the extension threaded bar. The pulling load was applied using the hydraulic hand pump with the required load increment and was measured through the calibrated hydraulic pressure gauge.

3.3 Test Procedure

The in situ pull tests were carried out in accordance with BS EN ISO 22477–5 (2018), and the loading cycle and reading frequency used in this paper were also in line with the

standard requirements. The selection of the loading cycles for the in situ pull-out test aims to measure the creep behaviour of the grouted rock bolt and assess pre-failure rock bolt movements. The maximum pull-out load (P_m) was considered to be 90% of the steel rebar's yield tensile strength for the investigation test (i.e. $P_m = 385$ kN), whereas the maximum test load for the acceptance tests was the design load of 130 kN. The investigation pull-out load was applied incrementally in six load cycles, from the datum load (P_d) (38.5 kN) (i.e. typically 10% of maximum test load (P_m)) to a maximum test load which was followed by unloading to the datum load for each cycle. The maximum loads for each cycle in the investigation test were $0.4P_m$ (154 kN) for cycle 1, $0.55P_m$ (192.5 kN) for cycle 2, $0.7P_m$ (269.5 kN) for cycle 3, $0.8P_m$ (308 kN) for cycle 4, $0.9P_m$ (346.5 kN) for cycle 5 and P_m (385 kN) for cycle 6 (BS EN ISO 22477–5 2018). The maximum load level for cycles 1 to 5 and cycle 6 was held for at least 15 and 50 min before unloading to assess any potential creep in the grouted rock bolt. For each load increment, the axial displacement of the bolt head was measured by the digital displacement gauge with an accuracy of 0.01 mm. In contrast to the investigation test, the pull-out load for acceptance tests was applied in one cycle. Three cases were proposed for the in situ pull-out tests, and their details, including the free and fixed lengths, maximum test load and type of test, are given in Table 2. To minimise the poor encapsulation along the bolt length (i.e. poor quality of grouting) and estimate the ultimate bond shear strength (grout bolt or grout rock), a shorter fixed length of 1 m was considered for the test cases I1 and I2 in the investigation

Fig. 3 In situ pull-out test setup

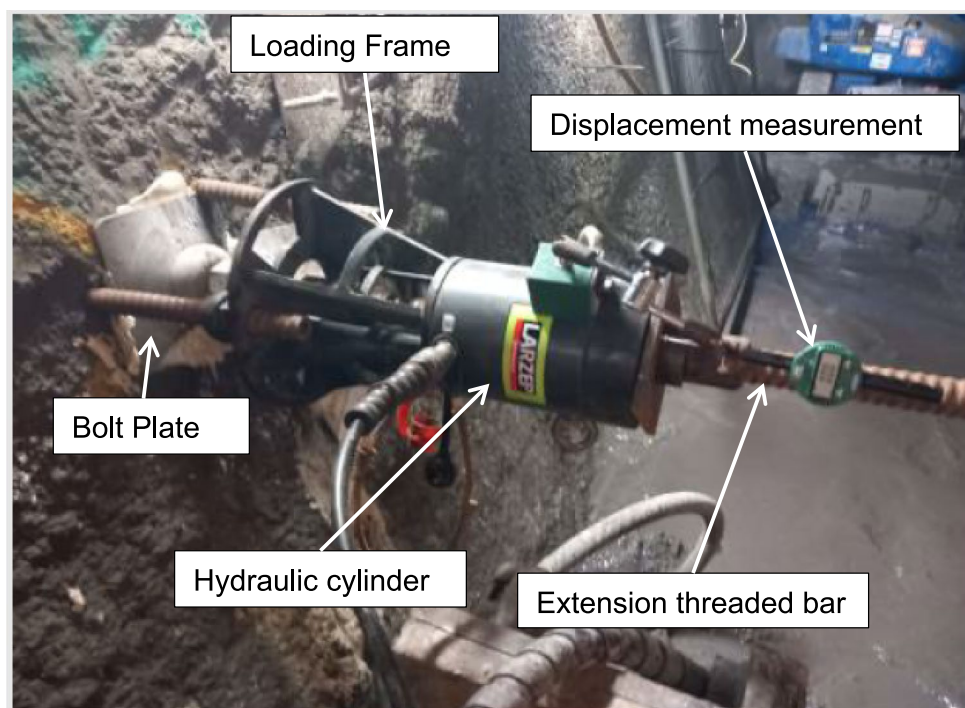
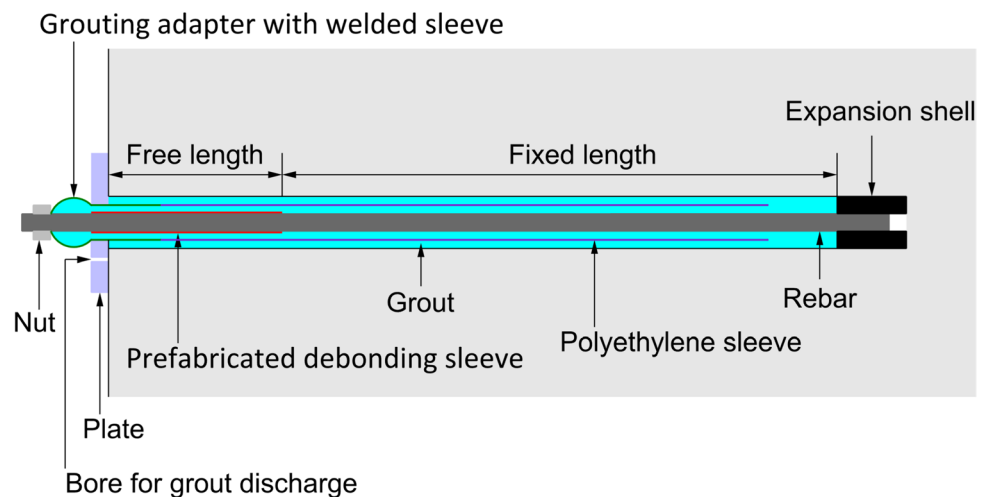


Table 2 In situ pull-out test cases and observations

Test case	Test type	Fixed Length (m)	Free length (m)	Borehole diameter (mm)	Proposed Maximum Test Load (kN)	Usage of mechanical expansion shell	Test observation
I1	Investigation	1.0	2.4	68	385	No	The grouted body slipped/failed during the third cycle of loading. During the third cycle, when pumping from 212 to 260 kN, the load dropped suddenly from 260 to 170 kN and the displacement increased from 14.83 mm to 22.58 mm. An attempt was made to continue the test but without success in maintaining the pressure
I2	Investigation	1.0	2.4	68	385	No	When the load reached 271 kN during the fourth cycle, the thread at the end of the bolt was damaged and a cracking sound was heard. The grouted body did not fail/ or slip
I3	Investigation	2.9	0.5	68	385	No	No slippage or bolt failure was observed
I4	Investigation	2.9	0.5	68	385	No	No slippage or bolt failure was observed
I5	Investigation	2.9	0.5	68	385	No	No slippage or bolt failure was observed
I6	Investigation	2.9	0.5	68	385	No	When the load reached 308 kN during the fifth cycle, displacement increased to over 42 mm. After the test, the bolt was visually pulled out. The bolt did not appear fully grouted (no grout was visible at the hole collar)
A1	Acceptance	3.5	0.5	68	130	Yes	No slippage or bolt failure was observed
A2	Acceptance	3.5	0.5	68	130	Yes	No slippage or bolt failure was observed
A3	Acceptance	3.5	0.5	68	130	Yes	No slippage or bolt failure was observed

tests. In addition, a longer fixed length of 2.9 m was considered for the other investigation test cases (I3 to I6) to assess the quality of grouting along the longer grouted length. The total length of the rock bolt used in all investigation tests was 3.4 m. Three more cases were proposed with the longer bond length and lower test load in the acceptance test. All the tests were conducted after the grout had attained one day of strength. For all test cases, the diameter of the drill hole

was 68 mm, which is the manufacturer's recommended size for the specific CT bolt used in the test. In addition, the drill hole diameter was kept constant to minimise the impact of drill hole size on the bond stiffness and strength. To achieve the required free length, a prefabricated additional internal sleeve with grease attached to each tested CT steel bar prevented bonding near the bolt head (see Fig. 4). The mechanical expansion shell in the CT bolt was not activated in any

Fig. 4 Modified CT rock bolt used in the pull-out test

of the investigation test cases (I1 to I6). In these test cases, the CT bolts were grouted passive rock bolts as shown in Table 2. The shell was only activated for the acceptance test cases (A1 to A3). In these tests, the maximum pre-tension applied was 10 kN, which was intended to delay grouting and minimise slack in the steel. The mixed grout was pumped through the grouting adapter, travelling through the inner annulus formed between the steel bar and polyethylene sleeve until it reached the bolt end. It then returned along the outer annulus between the polyethylene and the rock. The return of the grout from the bore in the plate indicates that the desired fixed length has been achieved. The tests were conducted at the side walls and the crown of the tunnel, maintaining a minimum distance of 1.5 m between

adjacent test locations. Testing conditions, including the type, strength, and mix of grout, as well as the strength and quality of the rock mass, were kept as consistent as practical for each test.

3.4 Test Results

The pull-out load–bolt head axial displacement curve from the in situ pull-out test case I4 is presented in Fig. 5. For test case I4, the fixed and free lengths were 2.9 m and 0.5 m, and the pull-out load was applied in six load cycles. Except for load cycle 1, the bolt was loaded from the datum load (38.5 kN) to the maximum cycle load and then unloaded to the datum load up to the proposed maximum test load (385 kN). Figure 6 shows a zoomed version of load cycle 3 for test case I4. As shown in Fig. 6, the load cycle had three distinguished stages:

- Stage 1: reloading (loading from datum load to the maximum previous cycle load),
- Stage 2: loading (the maximum previous cycle load to the current maximum cycle load) and
- Stage 3: unloading (from the current maximum cycle load to the datum load).

Every load cycle has recoverable (i.e., elastic) and unrecoverable (plastic) displacement at the end of unloading (Figs. 5 and 6). A similar in situ pull-out study on the grouted rock bolt in high-strength rock mass from Forbes et al. (2020) also observed both elastic and plastic displacement displacements during the loading and unloading

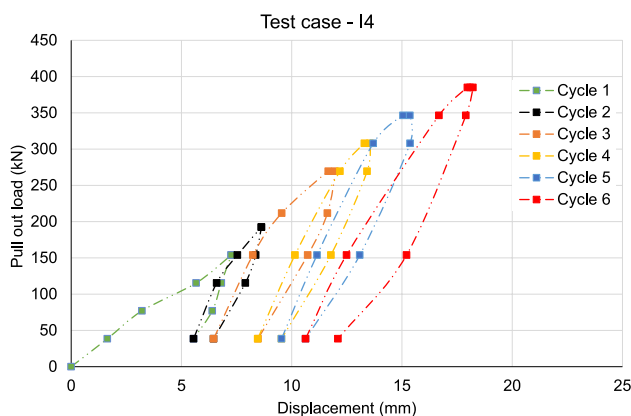
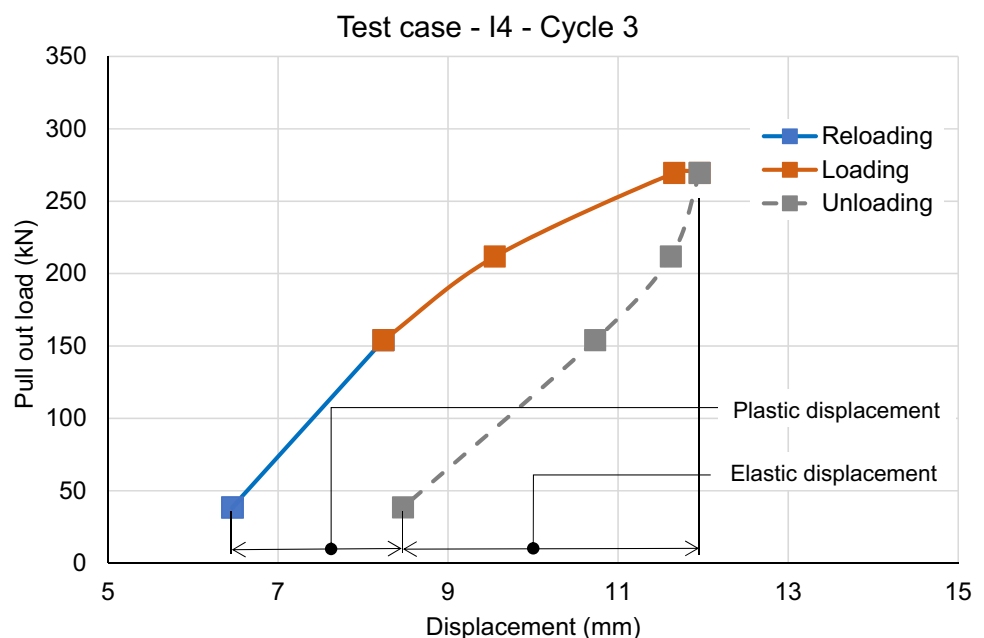


Fig. 5 Pull-out load–displacement curve for test case I4 (free length=0.5 m, fixed length=2.9 m, the maximum test load=385 kN, datum load=38.5 kN, no slippage or bolt failure observed)

Fig. 6 The zoomed version of the pull-out load–displacement curve of load cycle 3 of test case I4

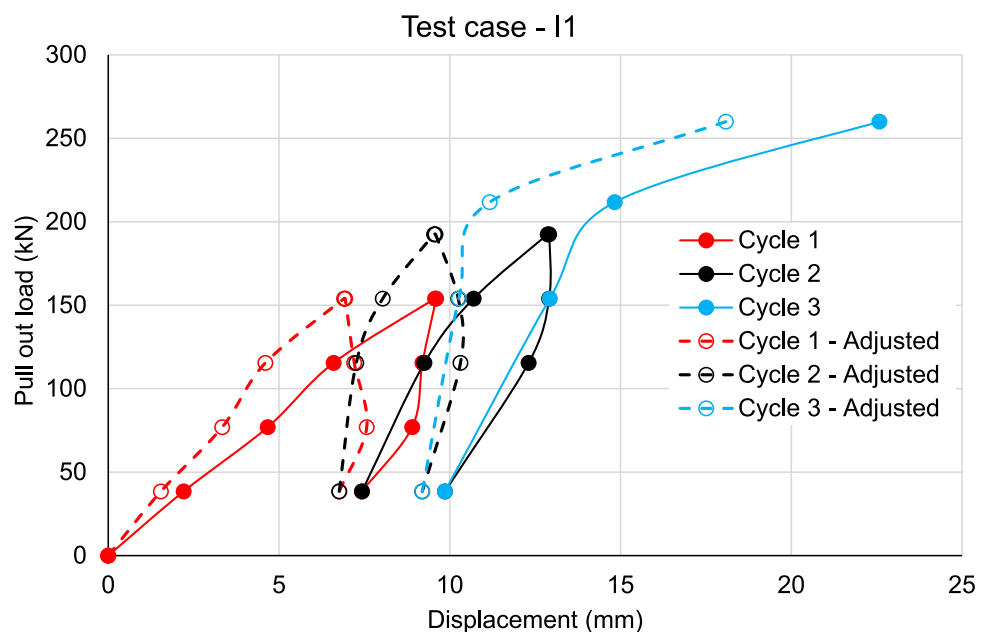


sequences, and it was concluded from their study that the plastic displacement was attributed to the compression of rock asperities at the interface during the pull-out test. For weak rock mass cases, the unrecoverable plastic displacement may be due to the shearing of minor asperities on the wall of the drill hole. In other words, the unrecoverable plastic deformation observed during the unloading stage of the pull-out test may be compared to a similar mechanism observed during the shear and joint closure testing on rock joints (for example, Bandis et al. 1983). The applied pull load may mobilise the interface between rock and grout during the loading and the contact of the minor asperities in the interface of rock and grout may increase and subsequently, some of the minor asperities undergo permanent deformation depending on the contact stress and strength of asperities. When unloading, the deformation of some of the minor asperities may cause a minor plastic or unrecovered displacement. The shear resistance from these minor asperities may not be significant compared to the cohesive strength provided by the bond between grout and rock. Therefore, the slippage of the grout column prominently governs the cohesive strength of the bond. The plastic displacement for the load cycle from 2 to 6 was comparable except for load cycle 1. The pull load–displacement curves show that the reloading stage was stiffer than the loading stage. Forbes et al. (2020) noted a similar behaviour during reloading. This reloading pattern may be typical of geomaterials confirming the radial deformation of the rock mass. This behaviour is similar to the behaviour of pile sockets in rock and rock-concrete joint shear behaviour, such that radial deformation, including dilation and compression upon shearing, governs the overall unloading–reloading behaviour (Ooi and Carter

1987). No bolt failure or bond slippage was observed in this case. No significant creep of the grouted rock bolt was observed during the period of sustained loading.

In contrast to test case I4, test case I1 had a shorter bonded length (1 m) with a longer free length of 2.4 m. As anticipated in the weak rock mass conditions, the grouted body of the rock bolt slipped during the third loading cycle for test case I1 (see Fig. 7). This observation confirmed that the predominant failure mechanism of grouted rock bolts in weak rock mass is expected to be the shear failure of the interface between rock and grout when the shear strength of the bond between grout and bolt is higher. The measurement of axial displacement at the bolt head encompasses both the bonded and unbonded sections of the rock bolt. To analyse the axial behaviour of the bonded portion, the induced axial displacement within the bonded portion should be differentiated. This can be achieved by subtracting the axial displacement of the unbonded portion from the total axial displacement measured at the bolt head (i.e., Axial displacement in bonded portion = Total axial displacement at bolt head — axial displacement in unbonded portion). The axial displacement of the unbonded portion is calculated using the formula for elastic axial deformation, which is: axial displacement of the unbonded rock bolt = $(\text{pull-out load} \times \text{unbonded length}) / (\text{cross-sectional area of the steel rebar} \times \text{Young's modulus of the steel rebar})$. The pull load–displacement curves for test case I1 were adjusted by excluding the elastic extension of the unbonded bolt length from the bolt head displacement (Fig. 7). The adjusted load–displacement curves show a slightly stiffer response and are comparable with the first two load cycles of test case I4 (a longer bonded length of 2.9 m with a shorter unbonded length of 0.5 m). The field

Fig. 7 Pull-out load–displacement curve for test case I1 (free length = 2.4 m, fixed length = 1.0 m, the maximum test load = 385 kN, datum load = 38.5 kN, grout–rock bond slippage observed when the pull-out load reached 260 kN during load cycle 3)



observations of other in situ pull-out test cases are given in Table 2. The pull-out load versus bolt head axial displacement curves for other test cases, such as I2, I3, I5, I6, A1, A2, and A3, are included in the Appendix.

4 Numerical Pull-Out test

As discussed in Sect. 3.4, the potential failure mechanism for the grouted CT bolt under a maximum pull-out load of 385 kN is expected to be a shear failure at the grout–rock interface especially in weak rock mass conditions. Therefore, the numerical modelling aims to calibrate the shear strength and stiffness of the interface between rock and grout by utilising the in situ pull-out test data. Two approaches can be used to numerically simulate the behaviour of grouted rock bolts under tensile load. The first approach explicitly models the actual geometry of the bolt and grout (Grasselli 2005; Jalalifar and Aziz 2012; Ho et al. 2019), which is computationally expensive. In contrast, the second approach uses equivalent structural elements, such as a cable (beam or truss elements) for the bolt and spring and a slider for the grout–rock or grout–bolt interfaces (Bin et al. 2012; Nemcik et al. 2014; Chen et al. 2018). Explicit numerical modelling requires several input parameters to simulate the constitutive behaviour of each element, such as grout, rock bolt, rock/host material, and interface. On the other hand, the equivalent structural elements numerical approach simplifies the mechanical behaviour with limited input parameters, which can be effectively back-calculated from the in situ pull-out test results (Bin et al. 2012). This approach's relatively quick modelling time makes it more practical for

large program-driven tunnelling projects. Therefore, this paper adopts the concept of grouted rock bolts based on the equivalent structural elements approach, as shown in Fig. 8.

4.1 Rock Bolt Modelling

As shown in Fig. 8, the rock bolt is modelled by a cable element divided into several nodal points with a defined spacing. The axial behaviour of the cable element is simulated using a one-dimensional constitutive model, which assumes many axial springs connected in series along the rock bolt (Itasca Consulting Group, Inc. 2019). The axial stiffness of each spring is calculated based on the bolt cross-sectional area (A_b), Young's modulus (E_b), and a spring element length or length between adjacent nodal points (L) and is given by:

$$K_{\text{bolt_spring}} = E_b A_b / L \quad (1)$$

The axial force–strain/displacement relationship for each spring element is given in Fig. 9. The induced axial force in the cable element is limited to the yield tensile capacity of the steel rebar (F_{t_yield}). After reaching the yield stage, the cable element can continue to carry axial force until the axial strain reaches the rupture limit ($\epsilon_{rupture}$). Once the axial strain reaches the failure limit, the cable element is considered to have failed and can no longer carry any axial force. All necessary input parameters (E_b , A_b , F_{t_yield} and $\epsilon_{rupture}$) to model the rock bolt element can be easily obtained from the rock bolt data sheet (Table 1).

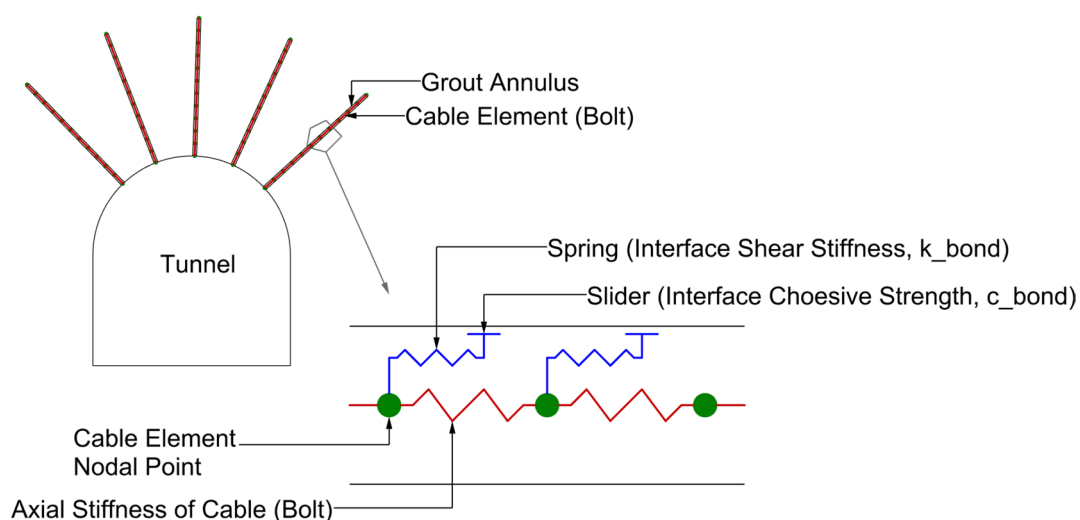


Fig. 8 Equivalent structural elements modelling approach for grouted rock bolt subject to tensile load (modified from Itasca Consulting Group, Inc., 2019)

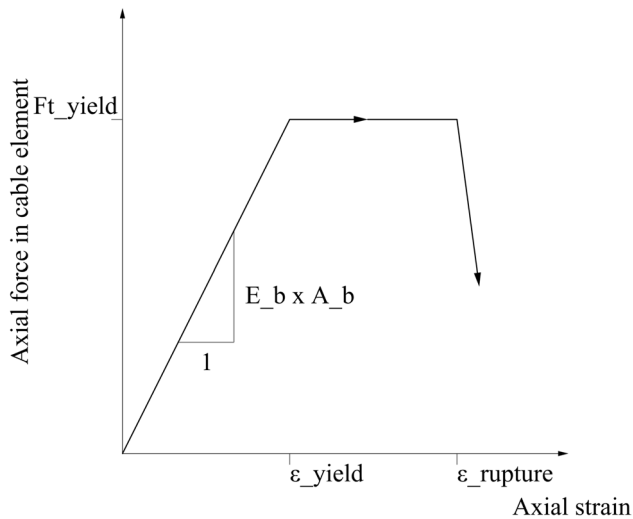


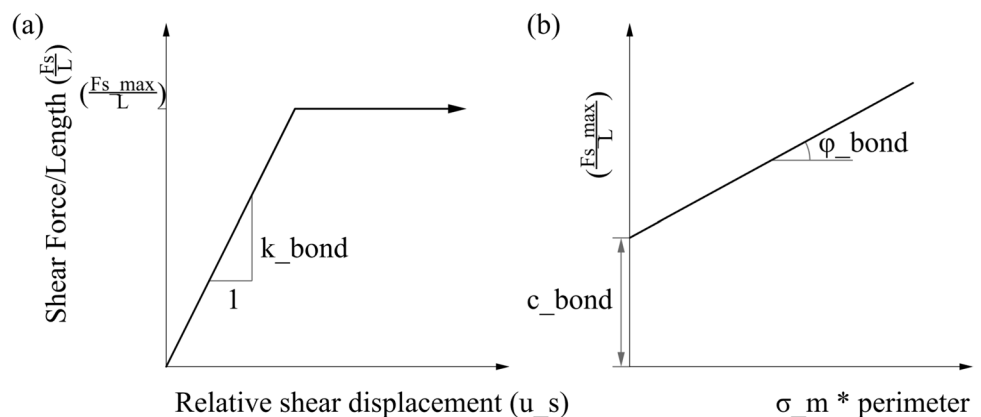
Fig. 9 Axial behaviour of cable element subject to tensile load (modified from Itasca Consulting Group, Inc., 2019). F_{t_yield} is the yield strength of the rebar, E_b and A_b are the elastic modulus and area of the rebar, ε_{yield} and $\varepsilon_{rupture}$ are the yield and rupture strains of the rebar

4.2 Interface Modelling

During the axial tensile loading of grouted rock bolts, the interfaces between bolt and grout and/or grout and rock are subject to bond shearing. The shear behaviour of these interfaces is expected to be cohesive and frictional. This behaviour can be numerically modelled as a combination of spring and slider connected to each nodal point, as shown in Figs. 8, 9 (Itasca Consulting Group, Inc. 2019). The mobilisation of shear force per unit cable element length along the interface with relative shear displacement is assumed to be linear until it reaches the peak/maximum shear strength (see Fig. 10a) and can be calculated by:

$$F_s/L = k_{bond} u_s \quad (2)$$

Fig. 10 The grouted rock–bolt interface mechanical behaviour: **a** shear force/length—displacement **b** shear-strength criterion (modified from Itasca Consulting Group, Inc., 2019). F_{s_max}/L is the maximum interface shear force per unit length, k_{bond} is the interface shear stiffness, c_{bond} and φ_{bond} are the cohesive and frictional strength of the interface and σ_m is the confinement around the bolt drill hole



where F_s is a shear force along the interface (grout bolt or grout rock), L is the cable element length between adjacent nodal points, k_{bond} is the interface shear stiffness (modelled as the shear spring), and u_s is the relative shear displacement between the cable nodal point and its relevant rock element point. The peak bond shear strength of the interface per unit cable element length is shown in Fig. 10b and defined by:

$$F_{s_maximum}/L = c_{bond} + \sigma_m \rho_{bond} \tan \varphi_{bond} \quad (3)$$

where $F_{s_maximum}$ is the maximum/peak shear strength of the interface, c_{bond} is the cohesive strength of the interface, φ_{bond} is the friction of the interface, σ_m is the confinement around the bolt drill hole, and ρ_{bond} is the perimeter of the potential sliding interface. If the rock and grout interface is the weakest, ρ_{bond} would be the perimeter of the drill hole; otherwise, ρ_{bond} would be the perimeter of the rock bolt. Nevertheless, the increase of interface normal stress due to shearing is expected to be minimal because the surface of the interface between rock and grout and grout and bolt are more likely to be smooth and planar, with no dilation (Indraratna et al. 2015; Thirukumaran and Indraratna 2016). Furthermore, for the testing programme subject of this paper, in situ pull-out test rock bolts were installed after completion of the tunnel excavation and support installation. At this stage, survey monitoring showed stabilisation of deformations, which meant that further induced stresses around the tunnel opening would not take place because full relaxation had been completed. In other words, a significant increase in normal stress along the drill hole was not expected at the end of the full relaxation stage. Therefore, it is reasonable not to consider the frictional component in the peak shear strength calculation as well as any confinement effect from the induced stress when modelling long-grouted rock bolts under tensile load only. As discussed in Sect. 2.2, the post-peak behaviour of the interface is assumed to be fully plastic (i.e. bi-linear slip model) for the long grouted rock bolt at in situ pull-out conditions (Fig. 10a). The required input parameters come down to the ultimate cohesive (bond) strength (c_{bond}) and

shear stiffness (k_{bond}) of the grout–rock interface for this current study. These parameters can be back-calculated from the numerical simulation of the in situ pull test results (Bin et al. 2012). As discussed in Sect. 3.4, the potential shear failure is expected at the grout–rock interface for weak rock mass conditions. A few equations were also proposed in the literature to calculate the shear stiffness of the grout–rock interface in the absence of in situ pull-out tests, and they are:

from Itasca Consulting Group, Inc. (2019);

$$k_{\text{bond}} \approx 2\pi G_g / 10 \ln(r_g/r_b) \quad (4)$$

from Cai et al. (2004);

$$k_{\text{bond}} \approx 2\pi G_g G_r / \left[(\ln(R/r_b) - 1/2) G_g + \ln(r_g/r_b) G_r \right] \quad (5)$$

and from Liu et al. (2017);

$$k_{\text{bond}} = \left[2\pi r_g \lambda_r G_g / \ln(r_g/r_b) \right] / \left[G_g / \ln(r_g/r_b) + r_g \lambda_r \right] \quad (6)$$

where G_g is the shear modulus of the grout, r_g is the radius of the grout borehole, r_b is the radius of the rock bolt, R is the influence radius of a single rock bolt (assumed to be $20r_g$ in this paper), G_r is the shear modulus of the rock mass, and λ_r is the shear stiffness of rock mass which may be assumed by one in ten times of normal or radial rock stiffness of the grout borehole cavity (i.e., $\lambda_r \approx 2G_r/10r_g$) (Ooi and Carter 1987).

In the absence of pullout test results, the shear strength of the grout–rock interface may be assumed to be 0.1 times the uniaxial compressive strength of the intact rock, σ_{ci} (Brown 2015). Therefore, the initial cohesive strength (c_{bond}) of the grout–rock interface can be calculated as

$0.1 \times \sigma_{ci} \times \pi \times D_{\text{borehole}}$, where D_{borehole} is the borehole diameter.

4.3 Model Calibration

4.3.1 Numerical Model

A series of numerical simulations were carried out using the 3DEC program developed by Itasca Consulting Group, Inc. (2019) to calibrate the in situ pull-out tests and results discussed in Sect. 3.0. The advantage of using the 3DEC program is its ability to simulate the realistic behaviour of grouted rock bolts under axial loading and continuous pull-out loading, similar to programs like FLAC3D. In addition, using the 'FISH' code as needed, the 3DEC program can easily incorporate various bond shear behaviours (such as bi-linear, tri-linear, and nonlinear models). The in situ tests conducted in this paper aimed to develop an in-tunnel lifting system supported by grouted CT rock bolts in a weak sedimentary rock mass (see Sect. 6 for application details). The 3DEC numerical program was utilised to analyse the lifting operation and its impact on the stability of the mined tunnel, as well as the existing tunnel support. Consequently, to maintain consistency in the design analysis of the lifting operation, the 3DEC was preferred for calibrating the relevant input parameters for the design of the grouted rock–bolt interface based on the pull-out tests. Figure 11 shows a typical 3D numerical pull-out test model. The model's width, length, and depth were $4 \text{ m} \times 4 \text{ m} \times 5 \text{ m}$. The host material, rock mass, was simulated as an elastoplastic material with the Hoek–Brown failure criterion. The input parameters for the rock mass are given in Table 3. These geotechnical input parameters were derived from a comprehensive geotechnical

Fig. 11 The 3D Numerical pull-out model

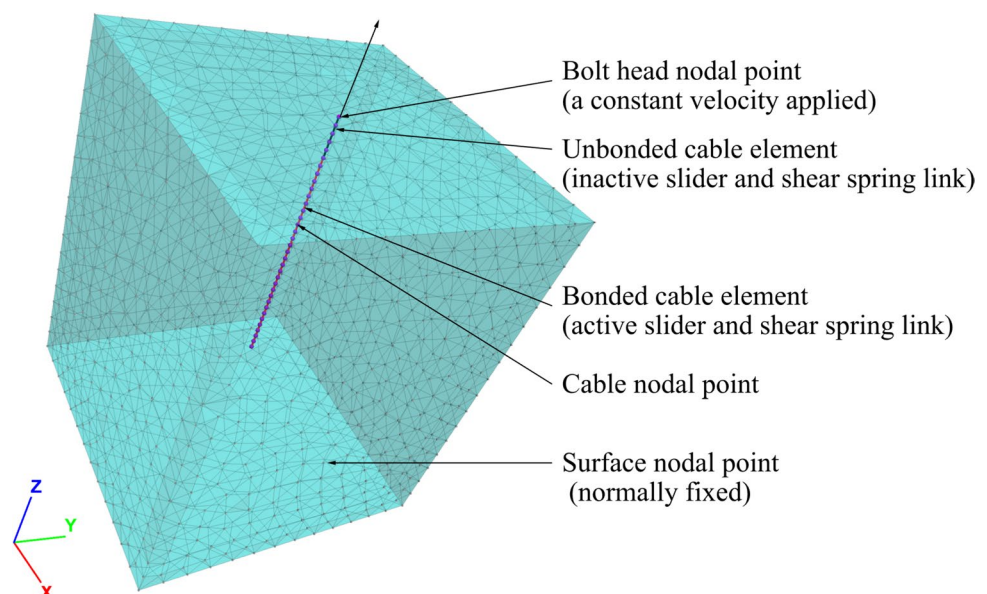


Table 3 Rock mass input parameters

Parameter	Value
Young's modulus, E_{rm} (GPa)	1.0
Poisson's ratio, ν_{rm}	0.3
Uniaxial compressive strength of intact material, σ_{ci} (MPa)	12
<i>Hoek–Brown model parameter</i>	
m_b	1.01
s	4.71×10^{-3}
a	0.504

interpretive report for the project, which included site investigations, laboratory and in situ testing, and mapping of the excavated tunnel face.

The rock bolt was modelled using cable elements with a spacing of adjacent nodal points of 0.1 m (referred to as L) (see Fig. 11). The axial behaviour of the cable elements is based on Fig. 9 and uses input parameters ($E_b = 200$ GPa, $A_b = 804$ mm², $F_{t_yield} = 416$ kN and $\epsilon_{rupture} = 7.5\%$) from Table 1. As discussed in Sect. 4.2, the spring and slider system was used to model the grout–rock interface for each nodal point. These springs and sliders were activated along the bonded length of the rock bolt (see red cable elements in Fig. 11) and not activated along the free length (see black cable elements in Fig. 11). The spring and slider system's input parameters include the grout–rock interface's shear stiffness and cohesive strength. An initial value of the shear stiffness of the grout–rock interface (k_{bond}) was calculated from Eqs. 4, 5, and 6 and used as an initial input for the simulation. The initial assumed shear strength of the grout–rock interface was 0.1 times the uniaxial compressive strength of intact rock, σ_{ci} (Brown 2015). These initial values of k_{bond} and c_{bond} were readjusted to calibrate final values for each in situ pull-out test result. This calibration will be discussed in Sect. 4.3.2. The surface element points of the 3D model were fixed. A constant velocity of 0.05 m/s was applied at the bolt head to simulate the in situ pull-out loading rate. The increment of axial load and displacement at the bolt head was measured for each cycle via a numerical code until a desired displacement or bolt failure was reached.

4.3.2 Calibration Results

The comparison between the in situ pull-out test and its numerical simulation results for the shorter or critical bonded length case I1 is depicted in Fig. 12. These results demonstrate the slippage between grout and rock, showing a good correlation. The calibrated grout–rock interface cohesive strength (c_{bond}) and shear stiffness (k_{bond}) were 260 kN/m and 21.5 kN/m/mm, respectively. The equivalent grout–rock interface bond stress is approximately 1.22 MPa

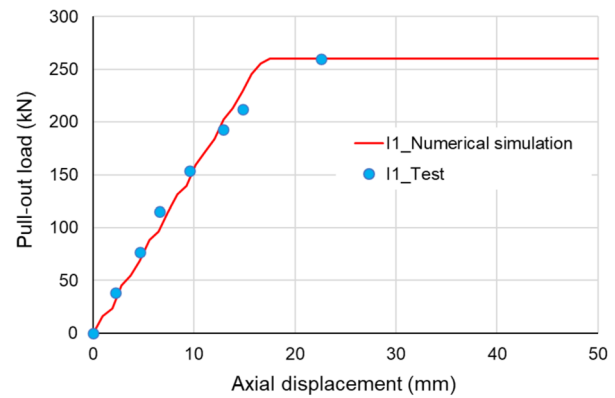


Fig. 12 Numerical calibration of the in situ pull-out test with the critical bond length (i.e. test case I1 and I2, bonded length = 1.0 m and unbonded length = 2.4 m, calibrated values of k_{bond} and c_{bond} are given in Table 4)

(i.e., 260 kN/m / π 68 mm), almost the assumed value of $0.1\sigma_{ci}$.

Figure 13 compares the in-situ pull-out test results and their numerical simulations for the longer bonded lengths with high pull-out load cases of I3, I4, I5 and I6. The calibrated grout–rock interface parameters k_{bond} and c_{bond} for these test cases are given in Table 4. The numerical simulations correlate well with the in situ pull-out test results for cases I4, I5 and I6 (see Fig. 13b, c and d). Among the numerical simulations of the test cases, test case I3 shows slightly less correlation with the in situ test (see Fig. 13a) than the other cases. For those tests without any failure, such as cases I3, I4, and I5, both in situ and numerical simulation results indicate a strain hardening behaviour after the pull-out load reached between 300 and 350 kN (see Fig. 13a, b, and c). As discussed in Table 2, the test case I6 experienced grout–rock interface failure or slippage before reaching the test load, with visible poor grout encapsulation. The test cases from I3 to I6 have similar bonded and unbonded lengths, borehole diameters, test load, grout mix and rock mass conditions. However, the initial response to the pull load (i.e., overall axial stiffness of grouted bolt system) before interface failure or the initiation of strain hardening varies among them. In other words, the calibrated grout–rock interface shear stiffness k_{bond} ranges from 6.5 kN/m/mm to 14.4 kN/m/mm, as presented in Table 4. This variation may impact the grouted rock bolt system's overall axial stiffness and pull-out response.

In addition, three additional in situ pull-out tests with much longer bonded lengths (4 m) and their numerical simulations are compared and presented in Fig. 14. Although test cases A1, A2 and A3 were conducted as part of production tests with a lower test load of 130 kN, the numerical simulation results closely match the in situ test results. The

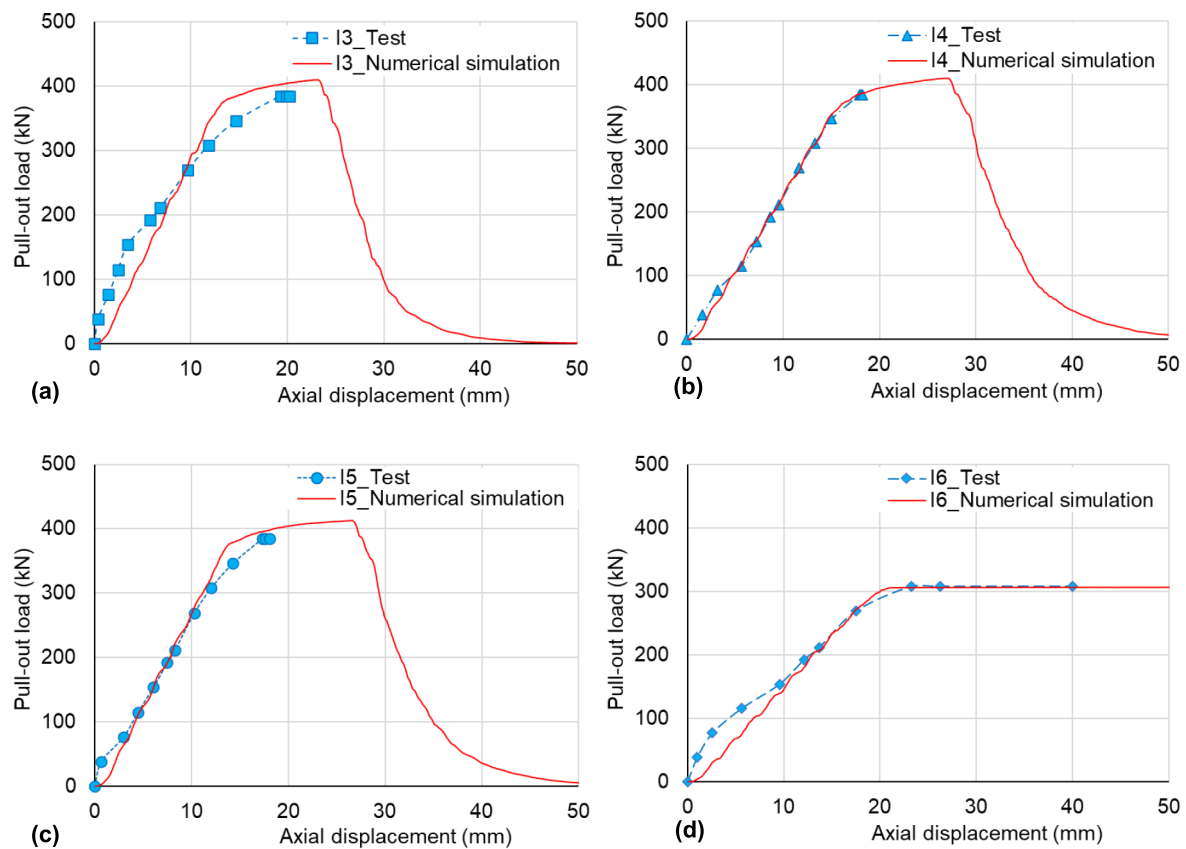


Fig. 13 Comparison between the in situ pull-out test results and their numerical simulations for the test series from I3 to I6 (bonded length = 2.9 m, unbonded length = 0.5 m, calibrated values of k_{bond} and c_{bond} are given in Table 4)

Table 4 Calibrated grout–rock interface parameters

Test case	Bonded length (m)	Unbonded length (m)	Calibrated grout–rock interface parameters				
			Bond cohesive strength, c_{bond} (kN/m)	Shear stiffness, k_{bond} (kN/m/mm)			
				From Eq. 4	From Eq. 5	From Eq. 6	From Numerical modelling
I1	1.0	2.4	260	119	452	344	21.5
I3	2.9	0.5	260	119	452	344	14.4
I4	2.9	0.5	260	119	452	344	10.8
I5	2.9	0.5	260	119	452	344	13.4
I6	2.9	0.5	106	119	452	344	6.5
A1	3.5	0.5	260	119	452	344	14.2
A2	3.5	0.5	260	119	452	344	7.1
A3	3.5	0.5	260	119	452	344	9.8

numerical predictions of these tests indicate that after the test load reaches 80% of the steel rebar yield tensile capacity, there is strain hardening before the rock bolt reaches tensile failure, similar to test cases I3 to I5. The numerical simulation curves show a strain-softening behaviour after

reaching the peak during numerical pull-out tests for investigation cases I3, I4, and I5 (Fig. 13) and acceptance cases A1, A2, and A3 (Fig. 14). However, the validation of this post-peak behaviour is limited due to the lack of in situ pull-out test data, including post-peak performance. The calibrated

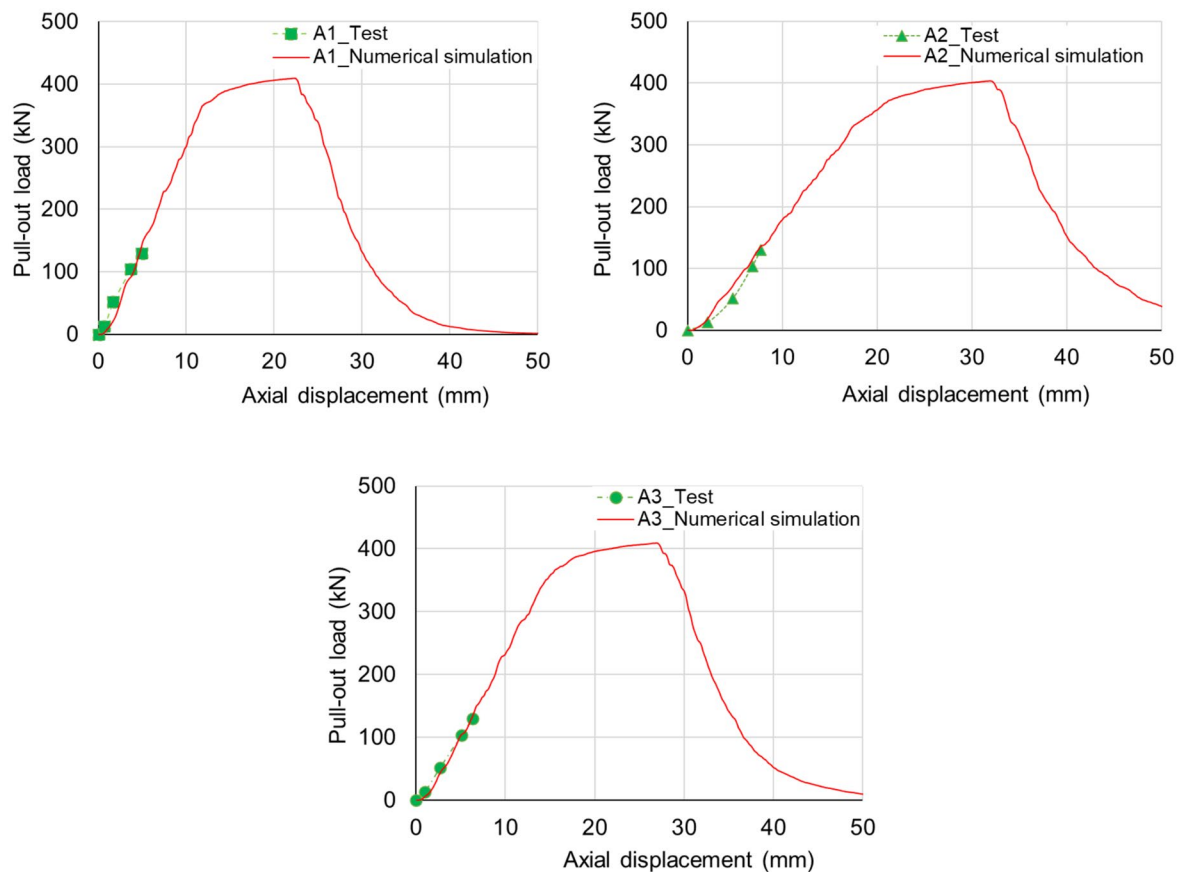


Fig. 14 Comparison between the in situ pull-out test results and their numerical simulations for the test series from A1 to A3 (bonded length = 3.5 m, unbonded length = 0.5 m, calibrated values of k_{bond} and c_{bond} are given in Table 4)

cohesive strength (c_{bond}) and shear stiffness (k_{bond}) of the grout-rock interface for these test cases are given in Table 4. The shear stiffness (k_{bond}) varies significantly among the calibrated interface parameters, ranging from 7.1 kN/m/mm to 14.2 kN/m/mm. This variation may be due to the quality of grouting along the bonded length. Villaescusa et al. (2008) and Ma et al. (2017) have reported similar findings in their studies on testing overcored grouted rock bolts.

Notably, the calibrated grout-rock interface shear stiffnesses (k_{bond}) obtained from the in situ pull-out tests ranged from 5 to 20% of the shear stiffness values calculated from Eqs. 4, 5 and 6 (Table 4). This significant difference is because, in addition to the effect of grouting quality, these equations do not account for the plastic sheath used in the CT bolt, which reduces the shear stiffness (Clarke and Sieders 2014). Therefore, using the k_{bond} values calculated from these Equations without comparing them to the in situ pull-out test results may result in incorrect rock-bolt performance. The shear stiffness of the grout-rock interface, calculated from Eq. 4, may be acceptable with an appropriate reduction factor between 0.05 and 0.2 to account for the

grouting quality and the presence of the plastic sheath in the CT bolt.

5 Assessment of the Quality of Rock Bolt Grouting

As discussed in Sects. 1, 2 and 4, in addition to the type, strength and mix condition of grout and strength and quality of rock mass, the grouting quality can impact the strength and stiffness of grouted rock-bolt interface in in situ conditions, mainly due to inadequate grout encapsulation. Existing testing on the field's overcored grouted longer rock bolts has suggested that poor grout encapsulation affects the stress mobilisation between the grout-rock interface (Villaescusa et al. 2008; Salcher and Bertuzzi 2018). In other words, poor grout encapsulation along the longer length of grouted rock bolts can reduce the cohesive strength (c_{bond}) and shear stiffness (k_{bond}) of the grout-rock interface. The post-production in situ pull-out tests are acceptance tests with a design or proof load much less than the yield or ultimate tensile capacity of the steel rebar. These tests may not

fully demonstrate the failure of the rock–bolt interface and the ultimate rock–bolt interface cohesive strength.

On the other hand, as discussed in Sect. 4.3, the interface shear stiffness can be calculated from pre and production in situ test results and their numerical calibrations. Furthermore, as discussed in Sect. 4.3.1, the numerical back analysis of in situ pull-out tests shows that the stiffness of the grout–rock interface (k_{bond}) varies significantly compared to the c_{bond} , indicating the quality of grouting. On the other hand, visually inspecting overcored longer-bonded rock bolts reveals inadequate encapsulation along the bolt length, particularly near the distal end of the bolt (Salcher and Bertuzzi 2018). In contrast, a shorter bonded length is expected to have a good grout encapsulation for a similar grout mix. Therefore, it seems reasonable to compare the interface shear stiffness of a shorter or critical bonded length with a longer bonded length to assess the relative quality of rock bolt grouting. This comparison assumes no significant variation in ground conditions along the different drill holes and no significant difference in overbreak along the drill holes. The calibrated interface shear stiffness (k_{bond}) values incorporate the effects of grouting quality and the plastic sheath. It is assumed that the impact of the plastic sheath on the interface shear stiffness (k_{bond}) does not vary between test cases, as the same material of plastic sheath is used for all test cases. Therefore, comparing the relative interface shear stiffness between critical and longer bonded lengths can cancel out the effect of the plastic sheath and highlight the impact of grouting quality. Based on this concept, the following equation could be proposed to estimate the relative quality of rock bolt grouting with available pre- and post-production in situ pull-out test results and their numerical calibration:

$$\text{Relative Quality of Grouting (RQoG)} = \left(1 - \frac{\text{Interface shear stiffness of critical bond length} - \text{Interface shear stiffness of longer bond length}}{\text{Interface shear stiffness of critical bond length}} \right) \times 100 \% \quad (7)$$

In Eq. 7, the interface shear stiffness of critical bond length can be obtained from the in situ pull-out test cases I1 to I2 and their numerical simulation as discussed in Sect. 4.3. Similarly, the interface shear stiffness for longer bonded length can be obtained from the test cases I3 to I6 and A1 to A3. Based on these interface shear stiffness values, the relative quality of grouting (RQoG) for longer bonded rock bolts can be calculated from Eq. 7. In the example calculation, for the longer bonded case I3, the numerically calibrated interface shear stiffness was 14.4 kN/m/mm, whereas for the shorter bonded case I1, it was 21.5 kN/m/mm according to Table 4, which was considered as the interface shear stiffness of the critical bond. Therefore, using Eq. 7, $\text{RQoG} = (1 - (21.5 - 14.4)/21.5) \times 100\% = (1 - 0.33) \times 100\% = 67\%$ for test

case I3. Similar calculations were performed for other longer bonded test cases, and their RQoG values are presented in Table 5.

Among test series on the longer bonded rock bolts (i.e., I3 to I6 and A1 to A3), test cases I3, I5, and A1 have shown relatively better grouting quality or encapsulation along the bonded length, with greater than 60% relative quality of grouting (RQoG) compared to the shorter or critical bond length. As expected, a lower RQoG percentage (30%) was calculated for test case I6, which failed along the grout–rock interface before reaching the test load, confirming the poor grouting quality along the bonded length. Similarly, the numerical simulation of test case A2 indicates poor grouting quality with a RQoG of 33%. However, no interface failure was observed during the in situ pull-out test for test case A2. This might be due to a longer bond length of 4 m and a maximum test load of 130 kN (proof load), which is less than the load required to cause the shear failure of the grout–rock interface. According to the standard acceptance criteria for a post-production test, test case A2 should be accepted because the pull-out load reached the test /proof load without any slippage or failure. However, based on the proposed method of grouting quality assessment, test case A2 could be reconsidered as acceptable due to having a lower percentage of RQoG, if reliance is placed on performance along the entire length of the bolt as a reinforcement element. In other instances, RQoG for test case I4 was 50%, with no failure or slippage observed. A similar percentage of RQoG was estimated for A3. Therefore, it is reasonable to consider test case A4 as an acceptable test. Based on the in situ tests and numerical simulations in this study (Sects. 3 and 4), an appropriate threshold value of RQoG is around

30%, which could be used as an additional acceptance criterion. However, it is essential to note that this threshold value was obtained based on limited tests and numerical simulations. Therefore, it should be reassessed based on each site or project specific. The applicability of this threshold value requires further validation with a similar weak rock mass before being put into practice.

The proposed quality assessment for longer rock bolt grouting is relatively quick and based on the load transfer mechanism of grouted rock bolts. In addition, the quality assessment of grouting provides an opportunity to review and improve the grouting mix, application, and installation within a short time frame. This verification ease is vital for program-driven large-scale projects.

6 Sensitivity of Grout–Rock Interface Parameters for an Application of Grouted Rock Bolts

6.1 Application

The application for which the rock bolts were tested involved a lifting system for a capacity of approximately 12 tons at a stabilised tunnel junction between a TBM and a mined tunnel. Due to space constraints, traditional heavy-lifting equipment was not considered. The proposed solution to this challenge involved a chain block lifting arrangement with six lifting points supported by grouted rock bolts and plates at each opening intersection (Fig. 15). During the lifting operation, four adjacent lifting points share the load. Each lifting point consists of two grouted rock bolts attached to a steel plate to support a working load of 8.8 tons. The design philosophy is to prescribe two rock bolts to work together and share the load at each lifting point. Suppose one of the rock bolts fails or is defective from installation (e.g. poor encapsulation and bond). In that case, the remaining rock bolt can support the total load, providing sufficient redundancy for unforeseen situations. The grouted CT rock bolts used for in situ pull-out tests and numerical simulations in Sects. 3 and 4 were utilised for the lifting point anchorage. Table 1 provides the mechanical and size properties of these bolts.

6.2 3DTunnel Model Simulation

To study the sensitivity of the grout–rock interface parameter, specifically the interface shear stiffness (k_{bond}) and the cohesive strength (c_{bond}) on the global response to the proposed lifting loading, a conceptual 3D numerical tunnel model was created using 3DEC (see Fig. 16). The model size was 40 m x 35 m x 5 m, consisting of a discontinuum zone of a jointed rock size of 20 m x 13 m x 5 m) and a continuum zone of the rock mass. The span and height of the tunnel were 11 m and 8 m, respectively. The geotechnical input parameters utilised in this study were based on the comprehensive geotechnical interpretive report for the project, which involved site investigations, laboratory and in situ testing, and mapping of the excavated tunnel face. The jointed rock was characterised by four sets of discontinuities with predominantly closed space

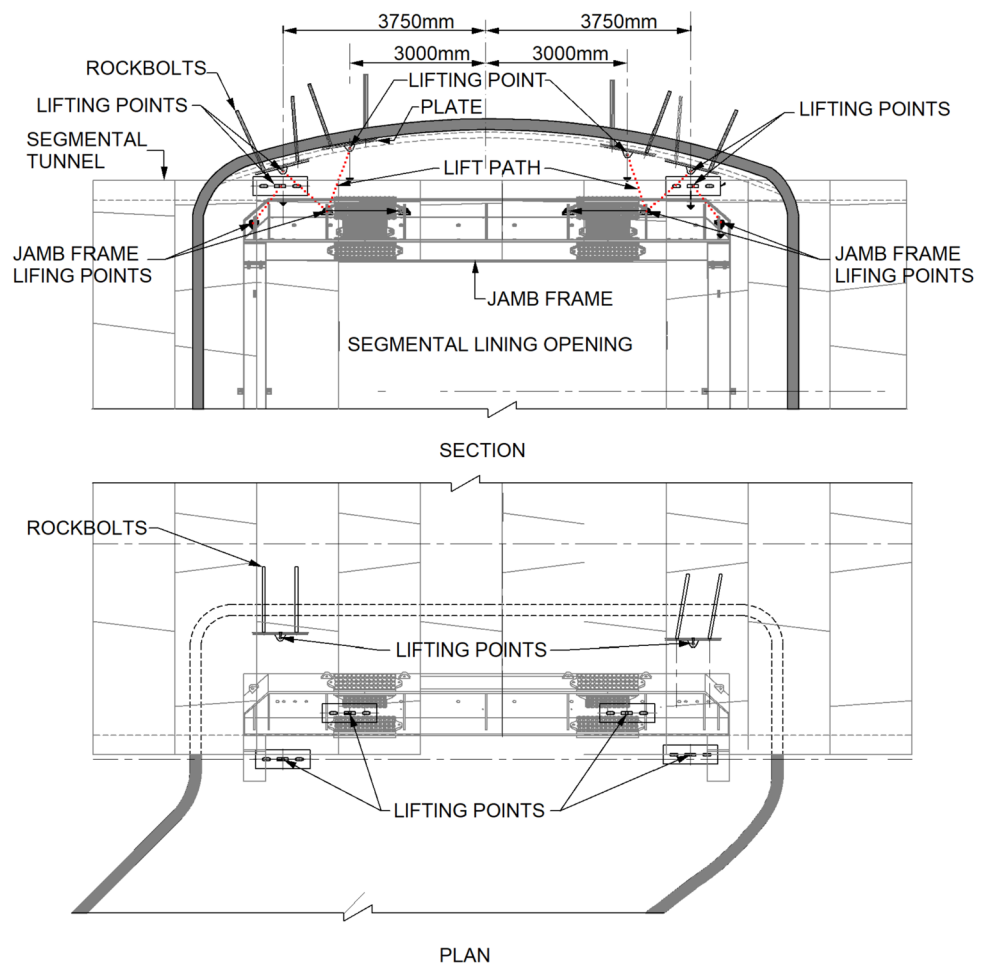
sub-horizontal bedding planes (average spacing = 0.3 m) and three sub-vertical joint sets (average spacing = 2 m, dip varies between 65 to 90 deg). The beddings and joints were modelled as an elastic-perfectly plastic Mohr–Coulomb model with friction of 27 deg and zero cohesion. A normal stiffness of 4GPa/m and shear stiffness of 0.4 GPa/m were assumed for bedding lanes, whereas the sub-vertical joints' normal and shear stiffness were 8GPa/m and 0.8GPa/m. The rock blocks in the jointed rock were simulated as an elastoplastic material with the Mohr–Coulomb failure criterion. The strength parameters for the rock blocks were assigned to cohesion of 1.2 MPa, friction angle of 45°, and tensile strength of 750 kPa. Rock block Young's Modulus was upgraded to 2 GPa with Poisson's ratio of 0.3 due to the inclusion of the discontinuities. The surrounding equivalent continuum rock mass to the jointed rock was assumed to be elastic with Young's modulus of 1 GPa and Poisson's ratio of 0.3. The in situ stress condition was considered to be hydrostatic (i.e. $\sigma_z = \sigma_x = \sigma_y$). The boundary condition for this 3D model assumed that the top face was free, while the normal velocity of all other faces was fixed.

The first modelling stage was to apply the boundary and in situ stress conditions and bring them to equilibrium. Then, the tunnel was excavated, and the tunnel relaxation was simulated by gradually reducing the initial stresses at the excavation wall to a target tunnel relaxation. In order to simulate 3D excavation sequences, a 70% tunnel relaxation was considered before installing the ground support. After the targeted tunnel relaxation was reached, the ground support was installed, comprising 3.0 m long Swellex-type bolts at 1.5 m x 1.5 m spacing, installed in a radial pattern to approximate the curvature expected in the true roof arch. A 50 mm thick shotcrete liner was applied to the crown of the tunnel to help retain wedges that may occur between rock bolts. The swellex bolt was modelled as a hybrid bolt element that simulates the dowel (i.e. reinforcement) and anchorage actions. The adopted hybrid bolt input parameters were a crosssectional area of $3.115 \times 10^{-4} \text{ m}^2$, young modulus of 1.98 GPa, yield strength of 240 kN, failure axial strain of 0.024, dowel shear strength of 12 kN, dowel shear stiffness of 26 MN/m, dowel shear failure strain of 0.4, bond shear stiffness of 50 kN/m/mm and bond strength of 230 kN/m. The supported tunnel was gradually relaxed to reach the equilibrium stage. The simulation results show that the

Table 5 The relative quality of grouting for the longer length of rock bolt

Test case	I3	I4	I5	I6	A1	A2	A3
Bonded length (m)	2.9	2.9	2.9	2.9	3.5	3.5	3.5
Unbonded length (m)	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Relative quality of grouting from Eq. 7 (%)	67	50	62	30	66	33	46

Fig. 15 An example of a jamb frame lifting arrangement for TBM segment lining opening at the intersection of TBM and mined outfall tunnels



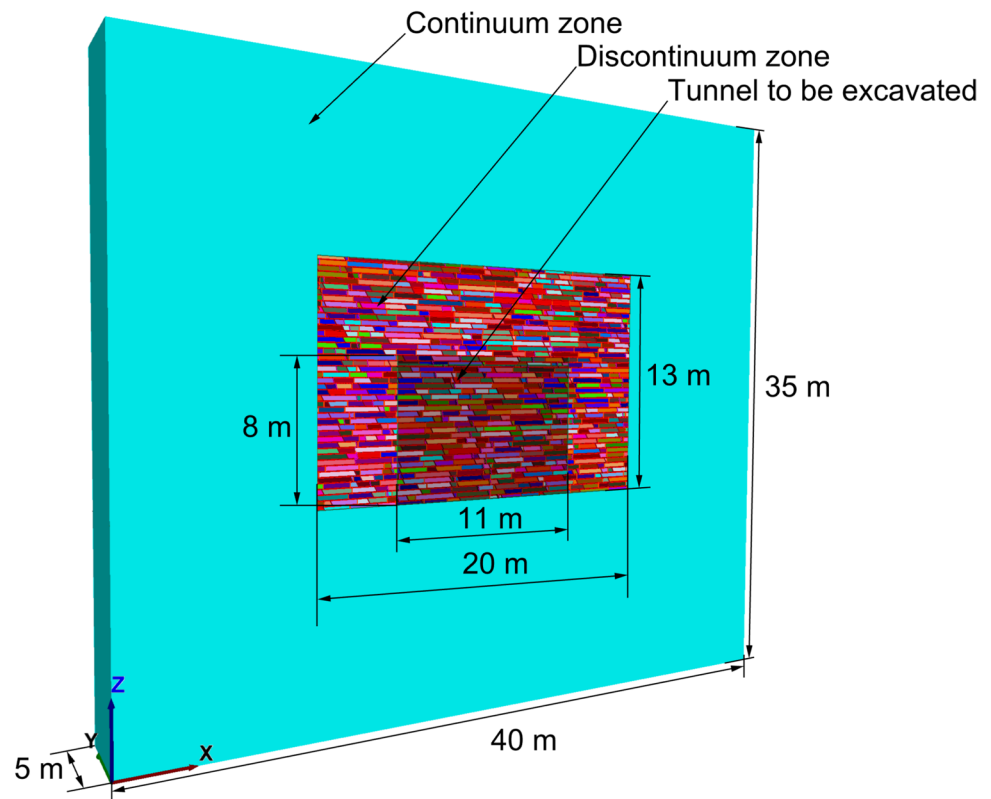
maximum induced vertical displacement of the mid-span was 36 mm, and the induced axial forces on the tunnel roof swellex bolts are within the yield strength (see Fig. 17).

6.3 Lifting Operation Simulation

As discussed in Sect. 6.1, the lifting operation was simulated with four adjacent lifting points. The first and second sets of lifting points were placed at 3.0 m and 3.75 m from the centerline of the tunnel, as shown in Fig. 18. For design purposes, one 4 m long CT rock bolt with 3.5 m of bonded (or fixed) length and 0.5 m of unbonded (or free) length was installed for each lifting point. As these grouted rock bolts provide the anchorage during the lifting, they were simulated with the cable elements. The input parameters for the grouted CT bolt modelling were identical to those adopted in Sect. 4.3. Two cases were considered to study the sensitivity of the grout–rock interface shear stiffness (k_{bond}) and the cohesive strength (c_{bond}). For case 1, c_{bond} was retained at a constant value of 260 kN/m and the values of k_{bond} were adopted from Eq. 4, Eq. 5, Eq. 6 and A1 and A2 in situ pull-out back analysis. For case

2, k_{bond} was kept at a constant value of 14.2 kN/m/mm, and the values of c_{bond} were 260 kN/m, 130 kN/m, and 100 kN/m. A constant velocity of 0.01 m/s was applied at each anchor point (i.e., bolt head) to simulate the gradual lifting of the load. The bolt axial load for each lifting point and the mid-tunnel span deflection were monitored during the lifting operation. Figure 19 shows the results from case 1. As shown in Fig. 19, for similar rock–grout bond strength, the bolt with a lower value of grout–rock interface shear stiffness (k_{bond}) requires more global deformation to transfer the load to rock mass compared to the bolt with a higher value of k_{bond} . Equations 4, 5 and 6 yield relatively higher k_{bond} values than the back calculation of the in situ pull-out tests due to the exclusion of the plastic sheath effect in these equations. The high values of k_{bond} may overestimate the CT grouted rock bolt performance under the proposed lifting operation. Therefore, a high-risk application such as lifting operations, in situ pull-out testing and numerical calibration should be required to estimate the realistic k_{bond} values and assess the bolt performance. As expected, for the given variation of grout interface bond strength (c_{bond}), there was no significant

Fig. 16 3DEC conceptual tunnel model



change to mid-span deflection during the lifting operation (see Fig. 20). Finally, the lifting operation has a negligible impact on the mid-span displacement, increasing the sag from 66 to 68.5 mm for critical cases (Fig. 19). The design predictions of mid-span deflection were within the measured displacement during the actual lifting operation.

7 Limitations

The proposed method for evaluating the quality of grouting has only been applied to CT rock bolts installed in a weak rock mass consisting of mudstone interbedded with limestone formation (with UCS of approximately 12 MPa and GSI of roughly 55) using a two-component (A and B) high-strength silicate resin. Furthermore, the diameter of the drill borehole was kept constant for all test cases to assess the grouting quality along the bolt's length. Varying drill hole diameters between tests could affect the bond shear strength and stiffness, and the accuracy of the proposed grouting quality assessment results. Consequently, it is essential to investigate the effects of different borehole sizes on the shear behaviour at the grout–rock interface, as well as the proposed grouting quality assessment, through in situ testing and numerical simulations. Therefore, before the proposal is taken into practical application in a more

general sense, it is necessary to determine if this method can be used for other rock formations, grout materials, bolting configurations, and a more extensive database.

This paper focuses on the slippage of grouted rock bolts at the interface between the rock and grout, a prominent failure mechanism observed in pull-out tests in weak rock masses. If similar pull-out tests are performed in relatively competent or high-strength rock formations, the assessment of grouting quality may be governed by the interaction between the grout and bolt, but similar changes in stiffness performance may be observed, which would also require further studies.

The equivalent element numerical modelling approach adopted in this paper, with reduced input parameters and relatively quick modelling time, makes it more practical to use in design practice. However, detailed numerical modelling, including all elements and their actual dimensions and mechanical behaviour (i.e., explicit numerical modelling), may provide additional insight into how poor encapsulation affects the overall interface shear stiffness and load transformation between the rock bolt and the rock.

8 Conclusions

The paper draws the following key conclusions:

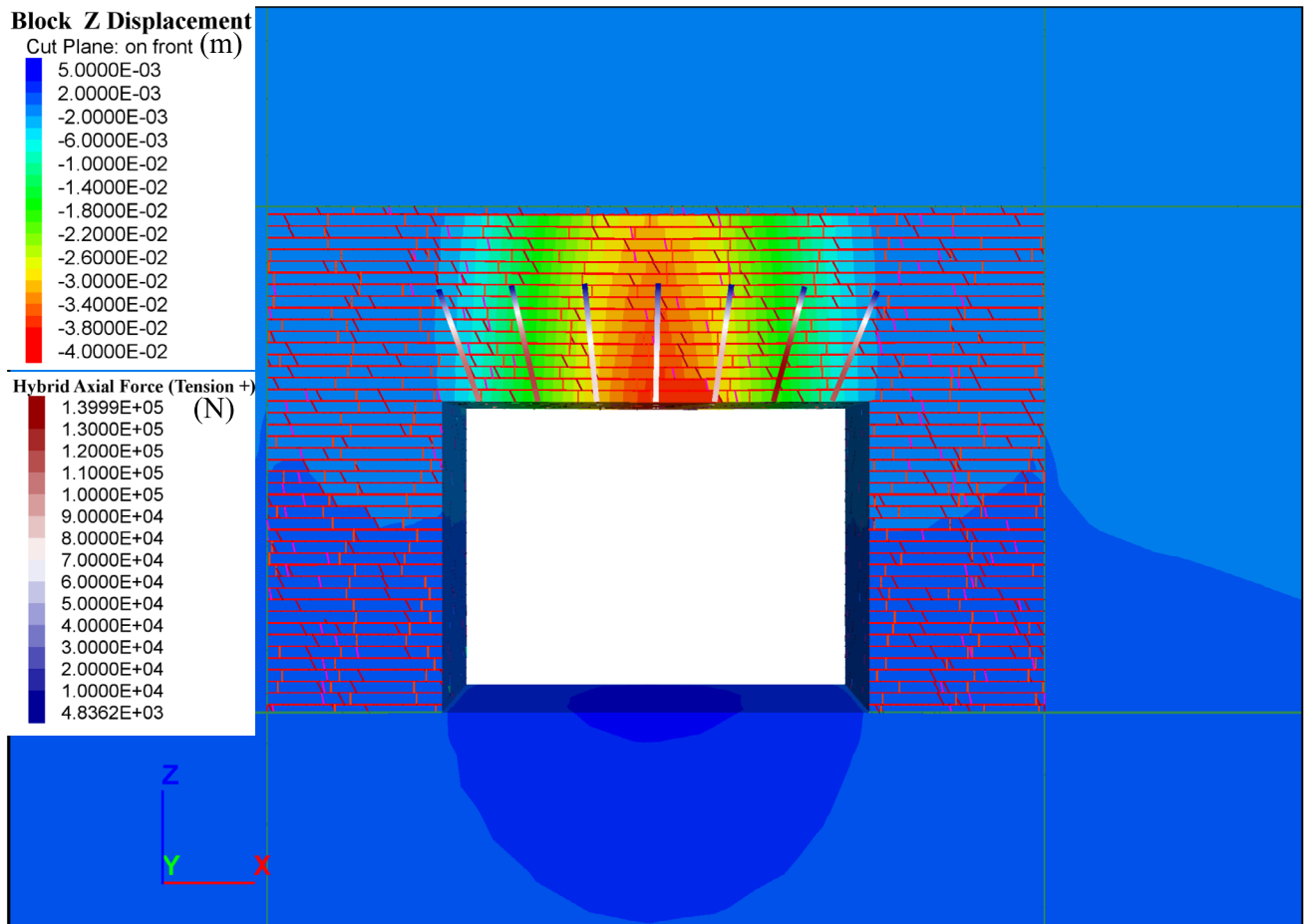


Fig. 17 The induced vertical tunnel displacement (in m) and axial forces (in N) on the roof swellex rock bolts after the equilibrium stage (i.e. before the lifting operation)

Fig. 18 The grouted CT bolt placement for each lifting anchor point (black—CT lifting anchor bolts, red—swellex bolts)

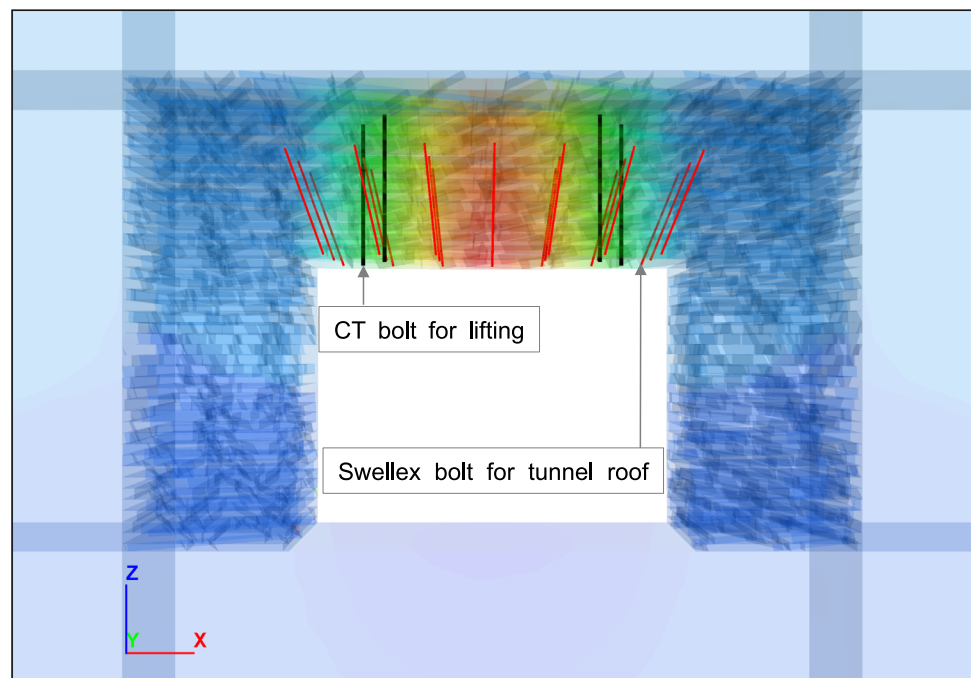


Fig. 19 Comparison tunnel mid-span deflection during the lifting operation with different values of the rock–grout interface shear stiffness (k_{bond}) while the interface cohesive strength (c_{bond}) has a constant value of 260 kN/m

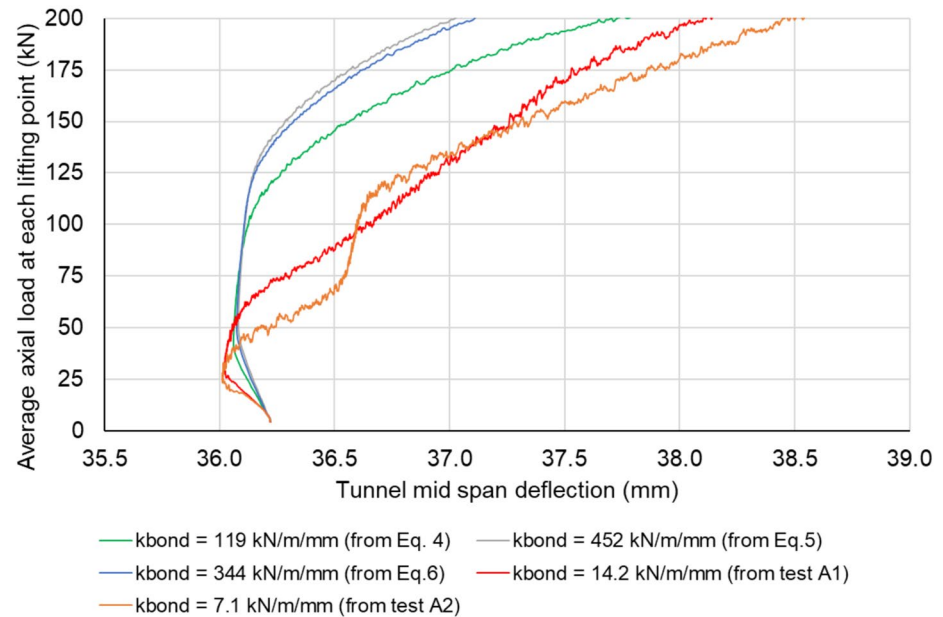
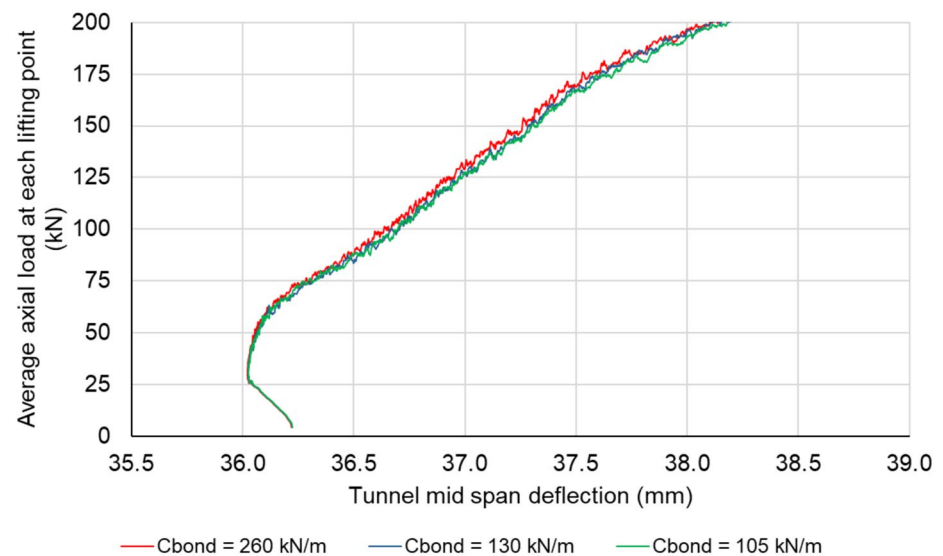


Fig. 20 Comparison tunnel mid-span deflection during the lifting operation with different values of the rock–grout interface cohesive strength (c_{bond}) while the interface shear stiffness (k_{bond}) keeps a constant value of 14.2 kN/m/mm



- **Modification of CT Bolt:** The standard CT bolt has been enhanced to include a free/unbonded length, improving its applicability for anchorage purposes.
- **In Situ Pull-Out Test Insights:** The results of the in situ pull-out test imply that in a weak rock mass such as the mudstone formation, the primary slippage of grouted rock bolts occurs between the rock and grout interface under the maximum test pull-out load of 385 kN. The numerical simulations correlate well with the in situ pull-out test results, showing the bond shear behaviour resembling a bi-linear model.
- **Variation in Shear Stiffness:** The numerical simulation results indicate that the shear stiffness of the grout–rock

- interface (k_{bond}) varies significantly among the calibrated interface parameters. It ranges from 6.5 kN/m/mm to 14.4 kN/m/mm for longer bonded rock bolts at the pre-peak stage, which likely reflects the variation in grouting quality along the bonded length.
- **Deviation in Calibrated Shear Stiffness:** The calibrated shear stiffness of the grout–rock interface (k_{bond}) obtained from in situ pull-out tests varied from approximately 5% to 20% of the k_{bond} values calculated using the closed-formed Eqs. 4, 5, and 6. This significant deviation between calibrated and calculated k_{bond} values may be attributed to variations in grouting quality along the different bolt lengths and the presence of the

plastic sheath, which is not accounted for in these equations. Consequently, relying on the k_{bond} values derived from these closed-form equations without considering the impact of grouting quality and double corrosion protection systems and comparing them to the in situ pull-out test results could result in incorrect rock bolt performance interpretation. Equation 4, among the closed-form equations, may be used to calculate the shear stiffness of the grout–rock interface. To account for the grouting quality and the presence of the plastic sheath in the CT bolt, an appropriate reduction factor between 0.05 and 0.2 should be applied.

- **Assessment of Grouting Quality:** The proposed Eq. 7 shows good potential for assessing the relative quality of grouting (RQoG) for pre- and post-production in situ pull-out test results of longer grouted bolts. The assessment results indicate that 70% of in situ pull tests met the test load without any failures, with a minimum of 45% of RQoG.
- **Minimum RQoG Threshold:** The corresponding value of the relative quality of grouting (RQoG) to visible pull-out test failure was approximately 30%. Therefore, the study considered 30% of the RQoG as a minimum threshold for an additional acceptance criterion. However, this threshold was derived from limited tests and numerical simulations conducted on one weak rock mass formation type. It should be re-evaluated based on specific site project test results and/or a more extensive study. When the assessed RQoG value is below the threshold, the grouting mix and application should be

reviewed and improved to meet or exceed the RQoG threshold value.

- **Parametric Study Findings:** The parametric study on the actual practical case shows that the grouted rock bolt with a lower grout–rock interface shear stiffness (k_{bond}) requires more global deformation to transfer the load to the rock mass when compared to the bolt with a higher value of k_{bond} for a given bond strength.
- **Practical Application and Feedback:** The proposed grouting quality assessment method focuses on quickly evaluating grouting quality based on the in situ pull-out tests and their numerical simulations. This fast assessment allows for prompt review and improvement of the grouting mix and application, which is crucial for large-scale projects. This approach has been put into practice for the specific project from which the tests were taken, resulting in a practical interpretation of the tests. This approach allowed adequate feedback to the construction team on the quality of the grouting operation, leading to the successful installation of the rock bolts and subsequent lifting operations for a large-scale energy project.

Appendix

The in situ pull test data for the investigation test cases I2, I3, I5, and I6, as well as the acceptance test cases A1, A2, and A3, are presented from Figs. 21, 22, 23, 24, 25 in this appendix.

Fig. 21 Pull-out load–displacement curve for test case I2

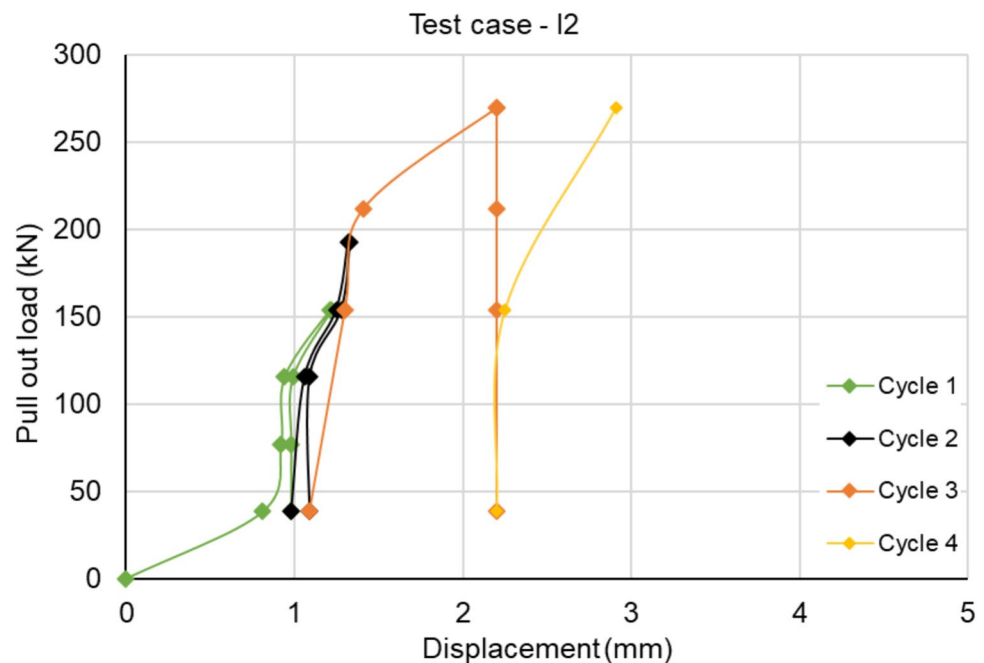


Fig. 22 Pull-out load–displacement curve for test case I3

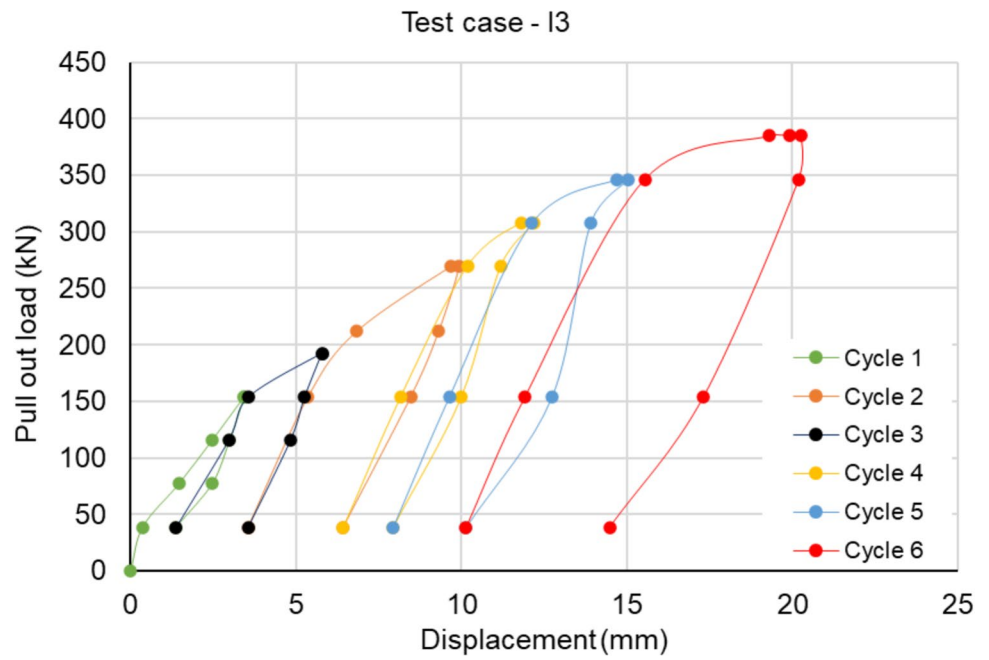


Fig. 23 Pull-out load–displacement curve for test case I5

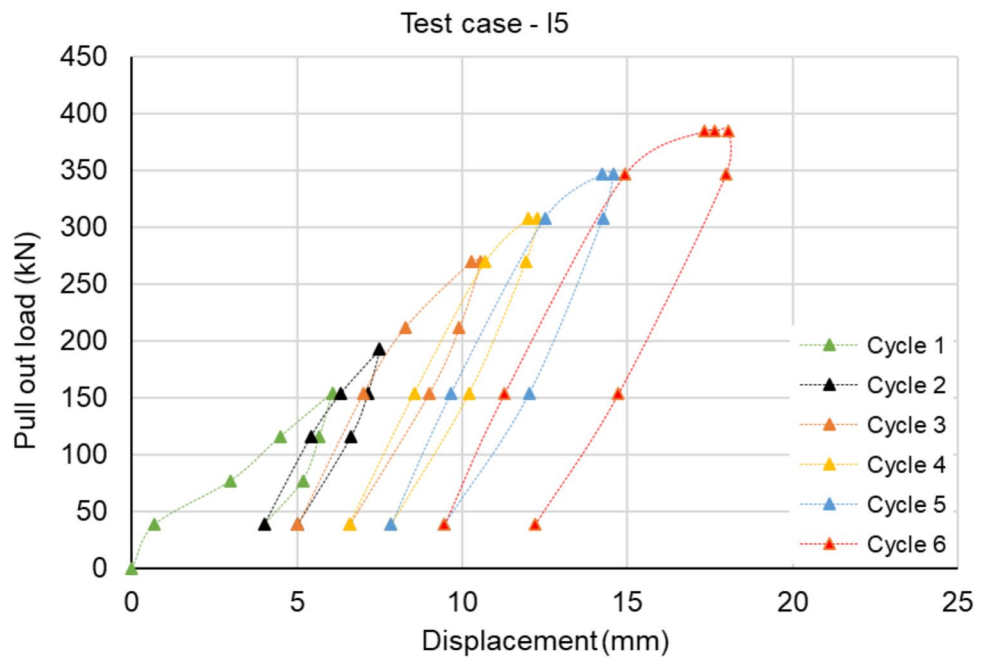


Fig. 24 Pull-out load–displacement curve for test case I6

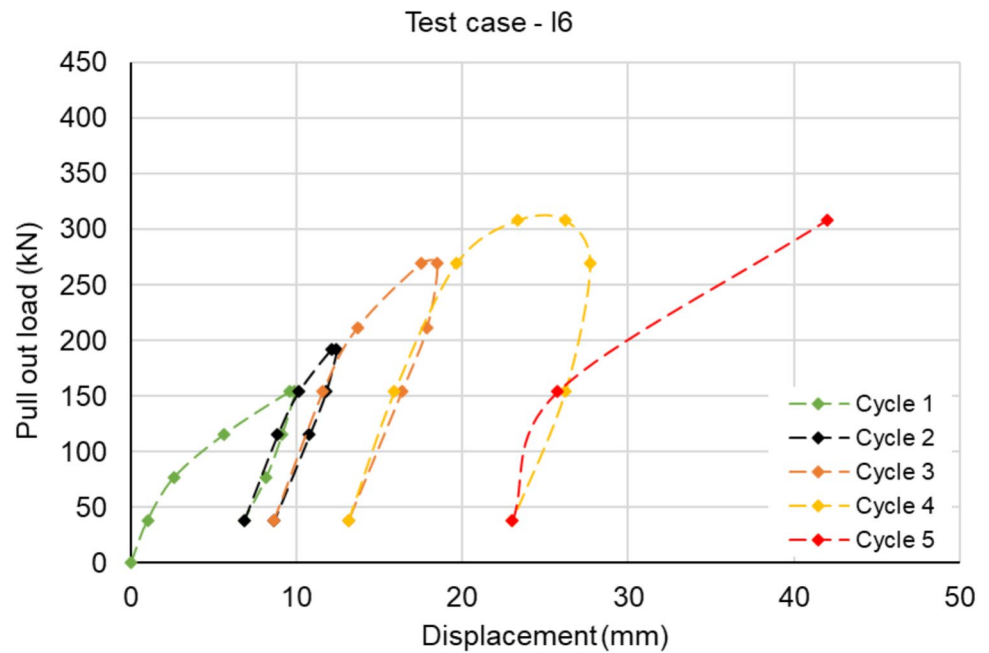
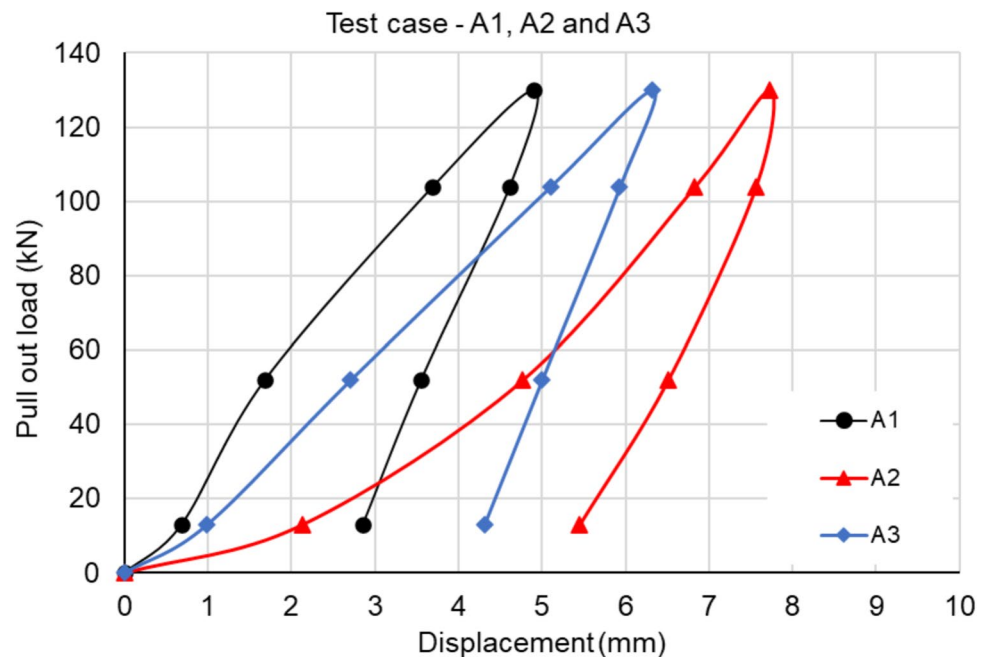


Fig. 25 Pull-out load–displacement curve for test cases A1, A2 and A3



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Author Contribution Statement Sivanathan Thirukumaran: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft. David Oliveira: Conceptualization, Writing – review & editing. Leo Ariza: Writing – review & editing.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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